



Infor Mongoose Multi-Site Implementation Guide

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Multi-Site Checklist

Follow this checklist when implementing a new multi-site system. Each step has a corresponding chapter that describes the step in detail.

Note: If you are using Infor Mongoose in the Cloud, ask your cloud administrator or consultant which steps apply to you.

Step	Description	Where
1.	Plan: • See the <i>Multi-Site Planning Guide</i> - Create spreadsheets and flowcharts. • Decide whether transactional or non-transactional replication will be used.	N/A
2.	Set up the database server: • Install Infor Mongoose and create one application database. • Depending on how you plan to set up your system, either create additional application databases for sites, or add sites to your existing application database. Use the Infor Mongoose Configuration Wizard. • If you are using one site per database, link the databases with Infor Mongoose Configuration Wizard.	Database server
3.	Set up the utility server: • Install and configure Infor Mongoose. • Create additional client configurations for other sites - Infor Mongoose Configuration Wizard. • Add web servers used by the configurations. • Define configuration groups. • Update Service Configuration Manager utility settings to: • Set up monitoring of configurations for events • Set up monitoring of configurations by TaskMan • Set up a replication configuration (non-transactional only) • Restart the services. • Use one console to administer all sites	Utility server

Step	Description	Where
4.	Set up end user clients.	Utility server or client machines
5. A	Set up transactional replication: • Set up replication links - Intranets form, Sites or Sites/Entities form. • Update views and triggers if linked sites are in different databases. • Set up replication categories and rules - Replication Categories form, Replication Rules form, Replication Management form. • Populate parameter _all records at other sites.	Sites
5. B	Set up master site and shared _all tables: • Set up transactional replication (above). • Set up the master site - Intranets form, Sites or Sites/Entities form, Intranet Shared Tables form.	Sites
5. C	Set up non-transactional replication: • Create replication user - Users form, User Modules form (requires license). • Set up system types - System Types form. • Set up replication links - Intranets form, Sites or Sites/Entities form. • Update views and triggers if linked sites are in different databases. • Set up replication categories and rules - Replication Categories form, Replication Rules form, Replication Management form. • Populate parameter _all records at other sites.	Sites
6.	Set up licensing: • If using intranet licensing, turn it on at the master site and any participating sites- Sites or Sites/Entities form. • License modules for a database on the License Management form	Sites
7.	Set up users and authorizations: • Set up sharing of user tables at the master site - Intranet Shared User Tables form (optional). • Set up parameters for passwords - Password Parameters form. • Set up other users - Users form, Object Authorizations for Users form, User Modules form (if using intranet licensing, run User Modules only at master site).	Sites
8.	Set up add-on products and modules.	Depends on the product

Follow these steps to plan your multi-site implementation:

- 1 See the *Multi-Site Planning Guide*. When you complete the steps in that guide, you will have spreadsheets and flowcharts containing most of the information you need to install and configure your sites. You can then update the spreadsheets to track changes as your system grows.
- 2 You must know which type of replication you are using before you set up your system, because some steps apply only to one type or the other:
 - *Non-transactional* replication, as used in this guide, refers to any inbound or outbound asynchronous replication. This can include applications that integrate with Mongoose by posting data via XML documents into the Mongoose sites, as well as Mongoose sites using XML to communicate with each other.
 - *Transactional* replication, as used here, generally means that sites are on the same database server and are replicating records through database triggers and stored procedures as soon as the records are added or changed on one site. The use of master sites and intranet shared tables requires much of the same setup as transactional replication.
- 3 Strongly consider setting up a test environment and making sure everything works properly before setting up your production environment.

Note: Be aware that this guide is intended for both cloud and on-premise versions of Mongoose. However, cloud users cannot perform all tasks that are described here. Cloud or hosted users should contact their business partner or Mongoose consultant regarding any task that involves access to the utility server or the database server.

Install Infor Mongoose and Create an Application Database with One Site

Note: The Configuration Wizard has an All-in-one database option that allows you to keep all of your forms, objects, and templates metadata in the same database as your application data. If you select that option, the references in this guide to the “Forms database” should be changed to the “Forms metadata tables in the application database.” References to the “Objects database” should be changed to the “Objects metadata tables in the application database.” The concepts are still the same. However, be aware that, if you use the all-in-one option for a database, you can only share forms, objects and templates metadata for sites that are in that application database.

On *each* SQL server machine where you will be adding site databases, follow the steps in the *Infor Mongoose Installation Guide* chapter “Setting Up the Database Server” to install the application and do the following:

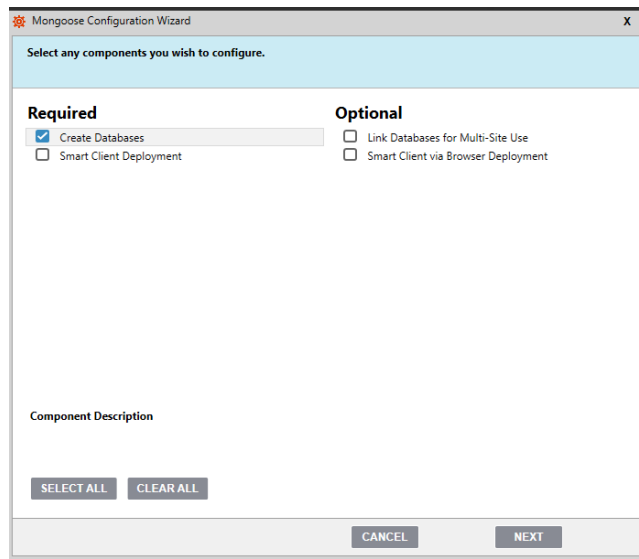
- Create one application database.
- Use your SiteEntity spreadsheet (from the Planning phase) to enter the site information when configuring the application database.
- Create at least one forms database for your system. Your SiteEntity planning spreadsheet should indicate how many forms databases you need and what to name each one. Later, on the utility server, you will specify which forms databases are used with which application databases.
- Create at least one objects database for your system. The SiteEntity planning spreadsheet should contain the information you need for this. Later, on the utility server, you will specify which objects databases to use with each application database.

After you create *one* application database, *one* forms database, and *one* objects database, return to this chapter to perform additional steps.

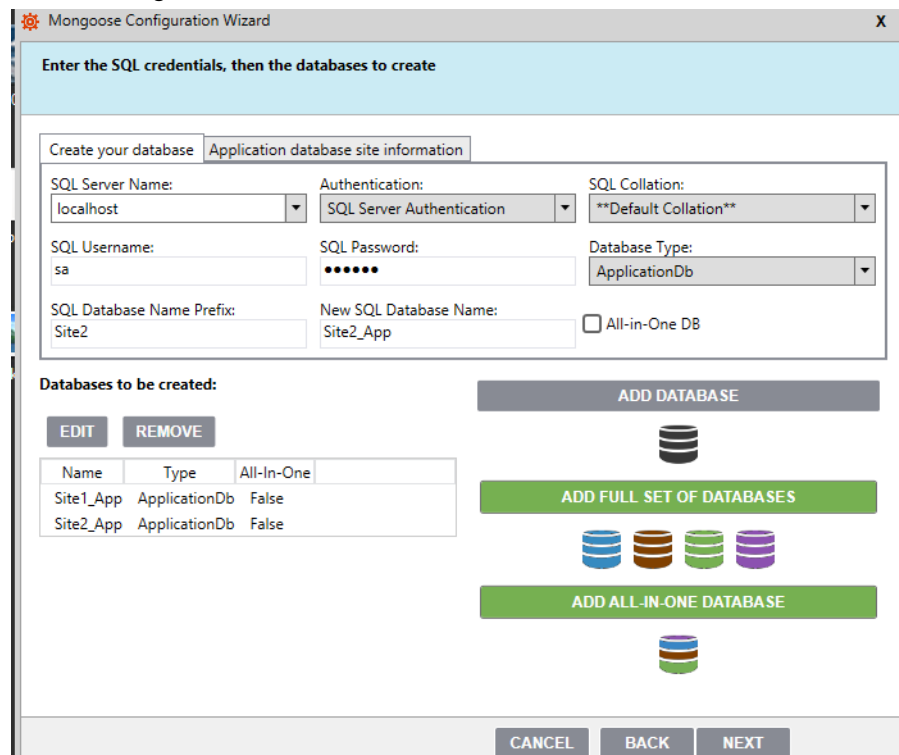
Create Additional Site Databases

To add multiple application databases on each database server, with one site in each database:

- 1 Run the Configuration Wizard again. (Run as Administrator.)
- 2 Select the **Advanced** option and click **Next**.
- 3 Select **Create Databases** and click **Next**.



- 4 You can add all of the application databases in one session. Just click the **Add Database** button after defining each database.



- a In the **Database Type** field, select **ApplicationDb**.
 - b To provide the database name and connection information, refer to your **SiteEntity** planning spreadsheet.
 - c Click **Next** to display summary information for each of the new databases.
- 5 Click **Commit** to commit the configuration changes. If you are creating several databases, this step may take some time to complete.
- 6 Click **Finish** to close the wizard.

Link Databases

Note: Perform the steps in this section only if you plan to use *transactional replication* to replicate data to or from sites in application databases located on this database server.

After you have defined all your databases, link them so they can communicate with each other. Follow these steps:

- 1 Run the Configuration Wizard again. (Run it as Administrator.)
- 2 Select the **Advanced** option and click **Next**.
- 3 Select **Link Databases for Multi-Site Use** and click **Next**.

Mongoose Configuration Wizard

Select any components you wish to configure.

Required ☐ Create Databases ☐ Smart Client Deployment	Optional ☒ Link Databases for Multi-Site Use ☐ Smart Client via Browser Deployment
--	--

Component Description
 Allows databases to be linked into multi-site configurations.

SELECT ALL CLEAR ALL

CANCEL NEXT

- Specify a **Group Name** for the set of sites on this machine that you want to link together. Use the group name from your SiteEntity spreadsheet.

Every time you run the Configuration Wizard, the first time you display this screen, the list of sites will be blank - even if linked databases already exist in the group you specify. To add a new site to an existing group, you must re-add all of the sites here. This sets up each database for multi-site communication and links it to all of the other databases that are added at the same time.

Note: Later, in the application's Site Groups form, you can add more groups or change the groups to which a site belongs.

- For *each* site that you want to be linked, specify this connection information for the database and click **Add**:
 - SQL Server Name:** Name of the SQL Server where the application database you want to link to resides.
 - SQL Username** and **Password:** Name and password of a SQL user who can access and update this database. Sometimes this is set up as the **sa** user.
 - Database to Link:** Enter or select the name of one of the application databases to be linked. (You will be adding the others in the next step.)
 - Site ID:** Select a site ID in the database you specified. For single-site databases, the site or entity ID displays.
 - Site Description:** the description of the database you specified is shown.

Note: Make sure you link only *application* databases (not forms or objects databases).

- When all of the sites you want to link are listed, click **Next**.
- Review the changes and click **Commit**.

- 8 Click **Finish** to close the wizard.

Note: Each time you run the Configuration Wizard on either the database server or the utility server, it appends information about the configuration changes to a log file found on that server:

Infor\Mongoose\Tools\ConfigWiz.log

- 9 If you plan to perform *transactional* replication *between sites on this database server and other sites that are not on this database server*, create a linked server definition in the SQL server master database of both the current site and the To site. Do this with the SQL Server Management Console: in the console tree, open the Server Objects folder for the server and right-click on the Linked Servers node.

This should match the list of linked servers you added in the Configuration Wizard.

Note: If all sites are on one database server, or if you are not using transactional replication, you can skip this step.

About Linking Databases

- Whenever you add a new site to an existing transactional multi-site environment, you must relink all sites using the Configuration Wizard. If you create and configure the new site by itself, it will prepare that database for multi-site work, but it will not know about other existing databases.
- Although the databases do not all have to be on the local server, transactional replication is much more efficient when they reside on the same server.
- You can run the wizard again to add more links later if needed.
- When you link multi-site databases with the Configuration Wizard, the following things are altered in the databases:
 - site table: new entries are added or updated.
 - *_all tables: row pointers are altered, foreign keys are temporarily dropped, tables are truncated and repopulated, and foreign keys are reapplied.
 - site_group table: new entries are added or updated.
 - site_live_link table: new entries are added or updated.
- If you plan to use *transactional replication* between all the sites in each intranet, but also use *non-transactional replication* between sites in different intranets, you should create a group for each intranet and link together the sites in each group. So you would run the wizard multiple times to create multiple linked groups.

Install Mongoose and Configure Components

On *each* utility/Web server machine where you will be creating configurations, follow all the steps in the *Infor Mongoose Installation Guide* chapter "Setting Up the Utility Server." These steps will:

- Install the application.
- Create an initial configuration on the utility server that links to *one* site database, using one of the configuration names you specified during your Planning phase.
- Set up the various Web services (IIS-based) on the utility server. These are used by clients connecting using http protocol, by the online help, for non-transactional replication, for integration with some applications, and so on.
- Set up Web rendering, if appropriate.
- Set up the TaskMan service for the utility server.
- Set up replication services on the utility server.

Note: Replication services are needed only if any of the configurations to be defined on this utility server will be running *non-transactional* replication.

- Set up Reporting Services so that you can print and preview reports.
- Set up the Application Search Web service, if appropriate.
- Through the Service Configuration Manager, set up monitoring of that first configuration by the TaskMan, replication, and event services.
- Set up an administrative client on the utility server that points to a specified URL. This URL is the utility/web server where the various configurations will be defined.

Then return to this chapter for the following additional steps to create configurations for the rest of your sites.

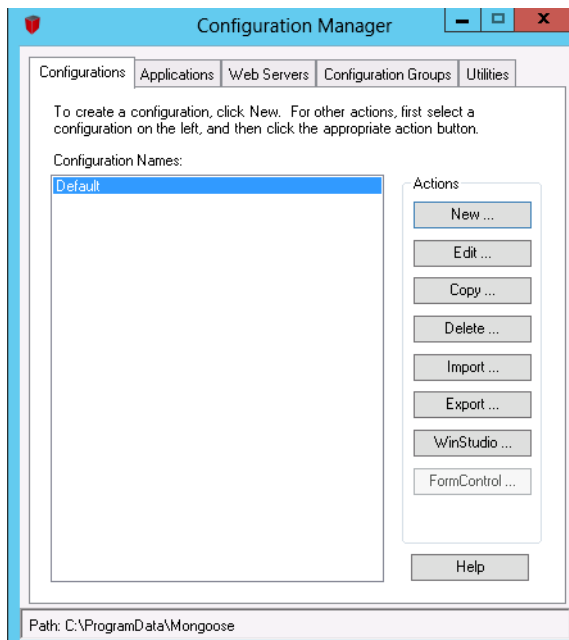
Create Additional Client Configurations

You must add a client configuration for *each* site database you want to access through this utility/web server.

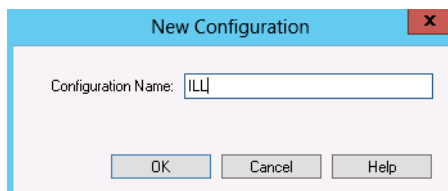
Caution: If you have multiple utility/web servers, only one of them should be designated as the Configuration Server, and the Configuration Manager utility should be run only on that Configuration Server. The resulting `MongooseConfigs.xml` file should then be copied to the other utility/web servers so they have access to the defined configurations. See the *Multi-Site Planning Guide* for information about the Configuration Server.

For *each* site, create one configuration name that exactly matches the site ID, including case, and that uses the site's application database. This is required unless you specify a different default **Configuration Name** on the Sites/Entities form. (You can create additional configurations for the site, with different names.)

- 1 On the utility/web server, open the Configuration Manager (Run as Administrator). For older operating systems, navigate to **Start > All Programs > Infor > Tools > Configuration Manager**. For newer operating systems, find Configuration Manager under Apps. The Configuration Manager displays the initial configuration that was created during installation and setup of the utility/web server.



- 2 Click **New** to create a new configuration.

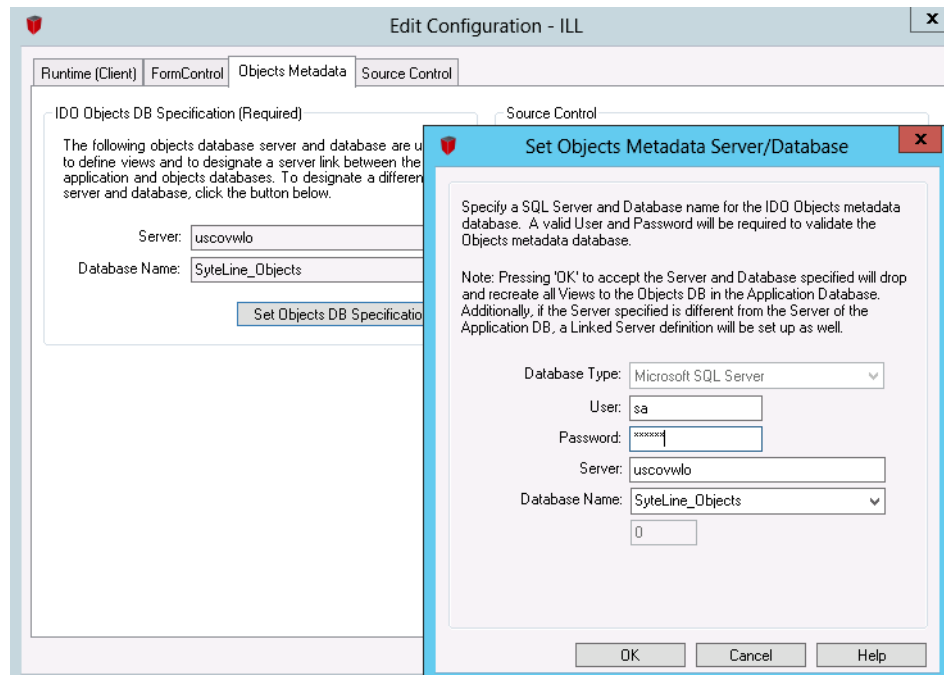


- 3 Enter the new configuration name and click **OK**.
- 4 In the **Runtime (Client)** tab, define the application and forms database connection information for the new configuration:

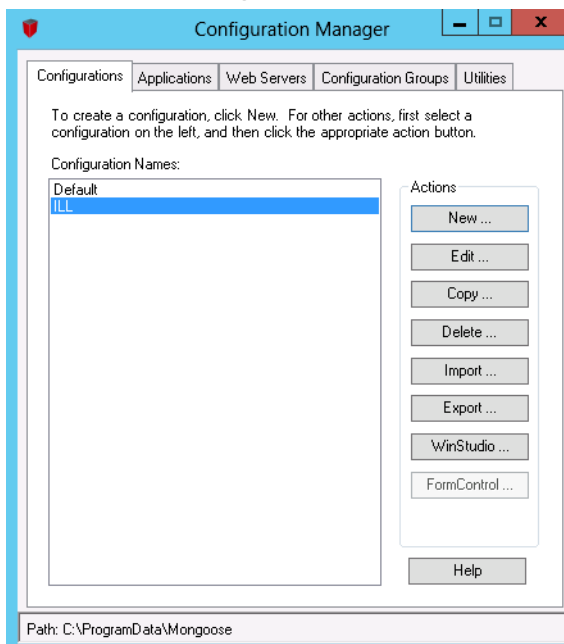
Note: Use the information from your SiteEntity and Configurations planning spreadsheets to help you fill in access information (SQL server name, login, password, and database name) for the application, forms, and objects databases.

- 5 In the **Objects Metadata** tab, click **Set Objects DB Specification** and then specify the Objects database name.

Note: You can associate only one Objects database with an application database.



- 6 Click **OK**. A confirmation message displays; click **OK** again.
- 7 If you use FormControl or a source control system, set up information on those tabs as needed.
- 8 Click the **OK** button at the bottom of the Edit Configuration screen to save the configuration and return to the **Configurations** tab, where the new configuration displays:



- 9 Repeat steps 2 - 8 until all configurations are defined.

Note: If a new configuration definition will be very similar to an existing one (for example, if just the application database name is different), you may want to use **Copy** instead of **New**. This creates a copy with a new configuration name and you then select **Edit** to modify the definition.

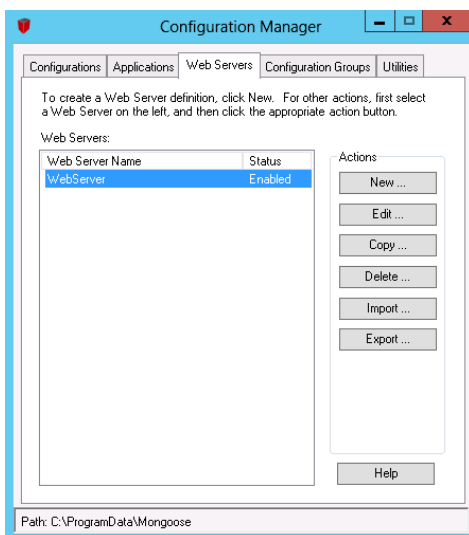
You must manually specify the Objects database in each copied configuration.

Add Web Servers Used by the Configurations

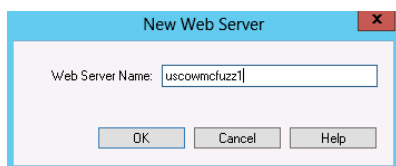
If you have multiple web/utility servers through which your clients can access the configurations, define them in the Web Servers tab of the Configuration Manager. They will be accessed in round-robin order as session requests come in to the Configuration Server. (See the **CAUTION** on page 16 about running the Configuration Manager utility only on the Configuration Server.)

Note: Do not include DMZ servers in this list.

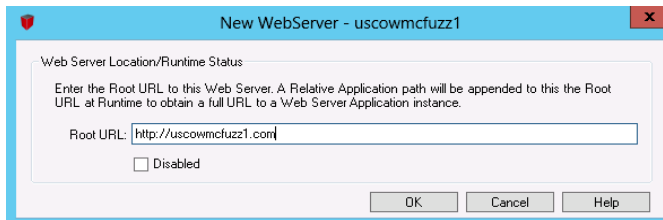
- 1 Click **New**.



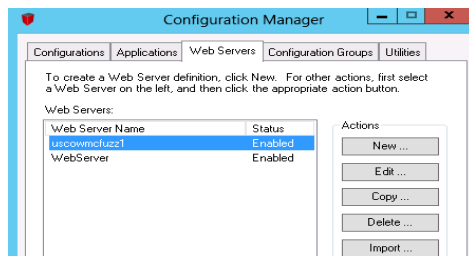
- 2 Enter a logical name for the web/utility server, and click **OK**.



- 3 Enter the http address of the web/utility server and click **OK**.

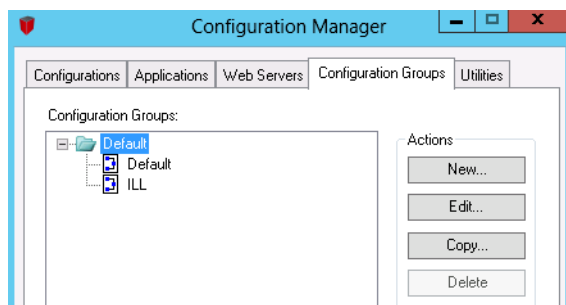


- 4 The server is added to the list of web servers the configurations can access.

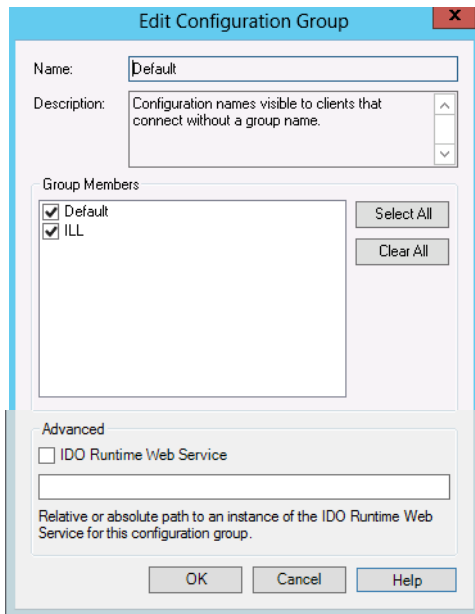


Define Configuration Groups

You should have determined during the Planning phase whether you will have multiple configuration groups. If so, select the **Configuration Groups** tab in the Configuration Manager utility.



Click **New** or **Edit** to display a dialog where you can define new groups and add configurations to them. The configurations must already be defined on the **Configurations** tab.



The screenshot shows the 'Edit Configuration Group' dialog box. It has a title bar with a close button. The 'Name' field is set to 'Default'. The 'Description' field contains the text 'Configuration names visible to clients that connect without a group name.' Below this is a 'Group Members' section with a list box containing 'Default' and 'ILL', both checked. To the right of the list box are 'Select All' and 'Clear All' buttons. At the bottom is an 'Advanced' section with a checkbox for 'IDO Runtime Web Service' and a text field for the path. Below the text field is a description: 'Relative or absolute path to an instance of the IDO Runtime Web Service for this configuration group.' At the very bottom are 'OK', 'Cancel', and 'Help' buttons.

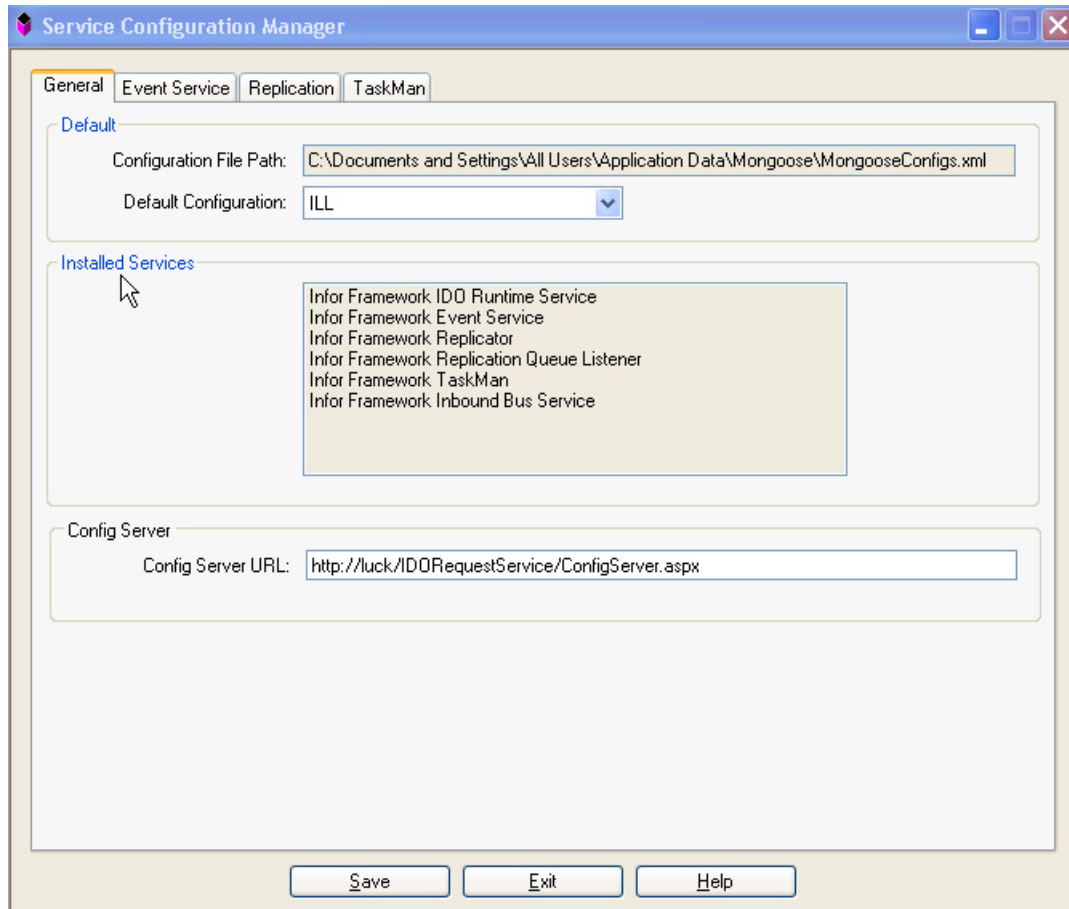
Configuration group names can contain only alphanumeric characters and the underscore character.

The Default group is predefined and cannot be deleted (but it can be edited). This group initially contains every configuration. If there are no configuration groups defined in your current configuration file (MongooseConfigs.xml), you will still see the Default group containing all configurations. Whenever you create a new configuration, it is added to the Default configuration group. If clients request configurations and do not specify a configuration group (or specifies "Default"), they will be presented with only configurations in the Default group.

Update Service Configuration Manager Settings

When you add configurations, you need to update the Service Configuration Manager utility so that the new configurations will be monitored by Infor services such as TaskMan, Replication, and the Event System.

Open the Service Configuration Manager on the utility server. (Run as Administrator.) For most operating systems, navigate to **Start > All Programs > Infor > Tools > Service Configuration Manager**. For Windows Server 2012, find Service Configuration Manager under Apps. .

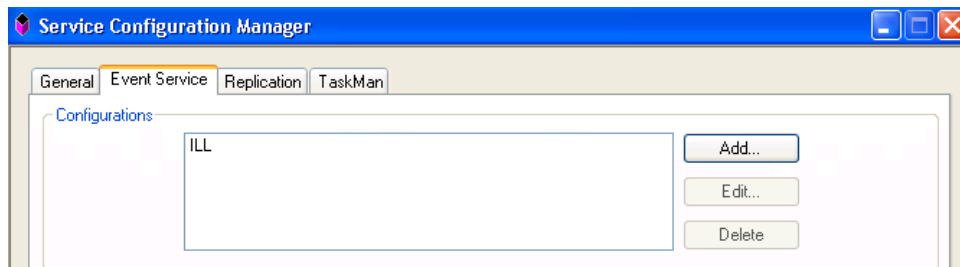


The **General** tab shows what services are installed on this utility server and where to find configuration files. You can also specify a default configuration here, but currently it only applies to the Replication services.

Use the **Config Server URL** field to specify the URL to the configuration server.

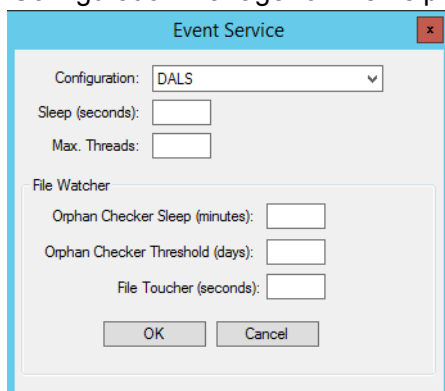
Set Up Monitoring of Events for Various Configurations

Use the Event Service tab on the Service Configuration Manager to specify each of the configurations you want the Event System to monitor.



See the *Multi-Site Planning Guide* for information about monitoring configurations for events. You may have created an Events spreadsheet as part of your planning process.

- 1 On the **Events** tab, click **Add** to display a dialog box where you can select a configuration and optionally specify sleep time (defaults to 10 seconds) and the maximum number of concurrent events that can run in this configuration (defaults to 5). For more information, see the Service Configuration Manager online help.



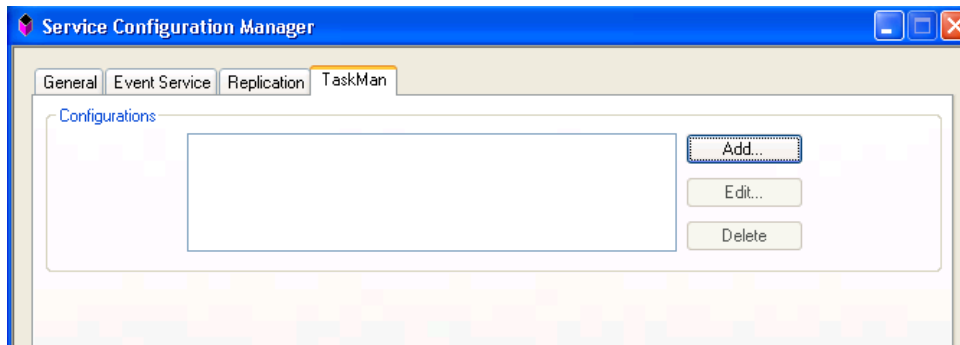
- 2 Repeat Step 1 for all configurations you want the Event System to monitor. Usually you will select all configurations for monitoring.
- 3 Click the **Save** button at the bottom of the window to save your changes. The system will not recognize the changes until you stop and restart the services later in this chapter.

Set Up TaskMan Monitoring of Configurations

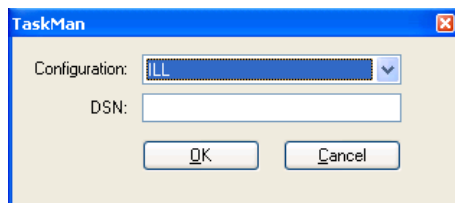
Set up TaskMan to monitor configurations for any application database on which users will be executing reports, stored procedures, IDO methods, executables, or other background tasks.

See the information on TaskMan monitoring of sites in the *Multi-Site Planning Guide*.

- 1 Go to the **TaskMan** tab of the Service Configuration Manager.



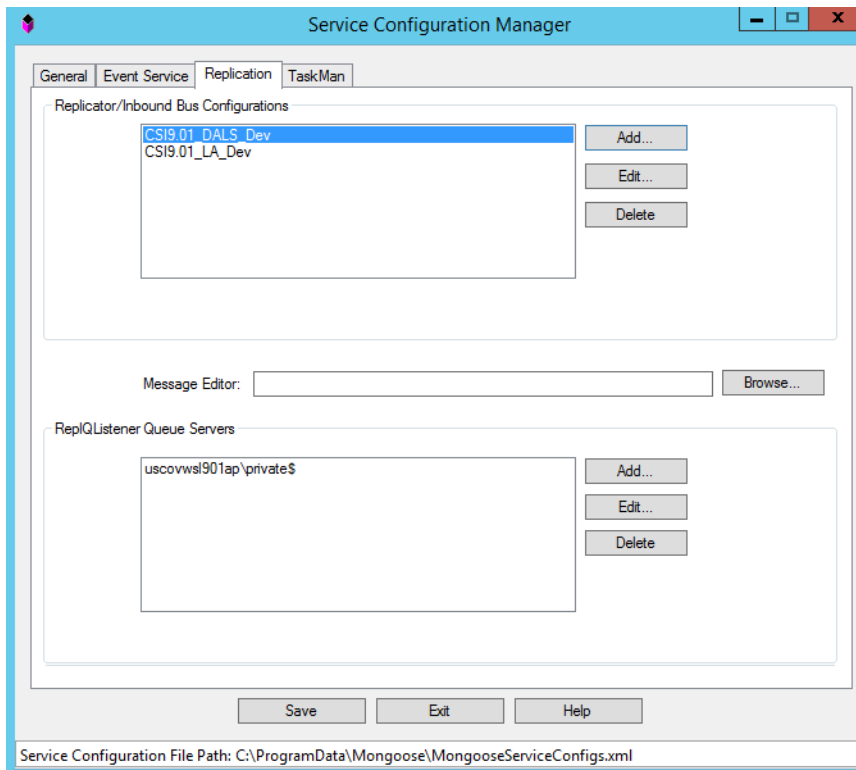
- 2 Click **Add** to display a dialog box where you can select a configuration and optionally a DSN. (DSNs are only needed for some custom reports created in previous versions. See the Service Configuration Manager online help for more information.)



- 3 Click **OK**. The configuration is added to the list in the TaskMan tab.
- 4 Repeat steps 2 and 3 until you have added all the configurations you want TaskMan to monitor.
- 5 Click the **Save** button at the bottom of the window to save your changes. The system will not recognize the changes until you stop and restart the services later in this chapter.

Set Up Replication and BOD Options

On the **Replication** tab, configure the options used in non-transactional replication and in BOD replication.



Refer to the online help for this tab of the Service Configuration Manager for details on how to set up these fields. Be aware of the following:

- If any of the configurations defined on this utility server will be running *non-transactional* replication (see the definition on page 7), click the **Add** button next to the **Replicator/Inbound Bus Configuration** field to create a list of all of the configurations that can receive inbound or outbound messages from other sites on the intranet. The Replication Document Outbox form at each site collects outbound BODs for that site. The Replication Document Inbox form at each site collects inbound BODs for that site.

Note: If you are upgrading from a previous version and you already have replication set up through a "bootstrap" site, it will still work as it did in the previous version. The **Replicator/Inbound Bus Configurations** field contains one site set as the Master Site, and the **ReplQListener Queue Servers** field has the Default field selected.

- The Inbound Bus Service can be installed multiple times on the same computer, in order to run it against a particular configuration that is specified at installation. However, setting a list of configurations to monitor, as described above, is the preferred method.
- If you set up a default configuration on the **General** tab, the system uses that configuration as the replicator configuration if you don't specify a different configuration here.
- Click the **Add** button next to the **ReplQListener Queue Servers** field to set up a queue server name for every intranet in your system.

Generally, the queue server name is set to `localhost\private$` which uses the standard private message queues named `inbound` and `outbound` on the utility server. However, if you have specified a Queue Server Name on the Intranets form, that name is used to build a path. For example, `utilityserver3\private$\MyQueues` results in the use of the queue names `MyQueues_inbound` and `MyQueues_outbound`.

- Click the **Save** button at the bottom of the window to save your changes. The system will not recognize the changes until you stop and restart the services later in this chapter.

Stop and Restart Infor Services

On the utility server, stop and restart the services so that they will be aware of any updates made in the Configuration Manager.

- 1 Go to **Administrative Tools > Services**.
- 2 Right-click on the **Infor Framework IDO Runtime Service** and select **Stop**. A message displays to tell you that stopping this service will also stop other dependent Framework services.
- 3 Click **OK**.
- 4 When the services are stopped, restart them again, starting the Infor Framework IDO Service first.

Note: It may be simpler to just restart the utility server machine. This automatically stops and starts all the services.

About the Services

Service Name	When to Stop/Start
Infor Framework Event Service Infor Framework Replication Queue Listener Infor Framework Replicator Infor Framework TaskMan	When you add, edit or delete configurations, or change which configurations are monitored, you must stop and restart these services so they recognize the changes. If you change the following information for a site, you must restart the <i>TaskMan service</i>: * Forms Database Name * Intranet Name * Polling Interval * Maximum Concurrent Tasks * Maximum Concurrent Report Tasks * Report URL * TaskMan Path If you change any of this information, you must restart the <i>replication services</i>: * the Intranet Name for a site * other non-descriptive information about the site or intranet * the password for a user who is set up in the Sites or Sites/Entities form for use in non-transactional replication.
Infor Framework IDO Runtime Service	Stopping the IDO Runtime Service also stops all dependent services (Event, Inbound Bus, Replication Queue Listener, Replicator, and TaskMan). When restarting the services manually, start the IDO Runtime Service first.
Infor Framework Inbound Bus Service	This service is not affected by configuration changes. The Inbound Bus Service now looks at the same configuration list as the Replicator Service.
Infor Framework Fax Service	This service, which monitors certain folders and picks up fax control files at specified polling intervals, is not affected by configuration changes.

Use One Console to Administer All Sites

You should be able to log in to client configurations for each site from a single console. This will simplify the setup and administration of your multi-site system.

If some of your sites are on a remote utility server, set up a remote client configuration as described in the *Infor Mongoose Installation Guide* chapter on setting up clients.

Install and configure end user clients as described in the *Infor Mongoose Installation Guide* chapter on setting up a client workstation.

Stop before you perform the steps in the "Run Infor Mongoose" section of that chapter, and return to this guide. Multi-site setup has some different steps - and many additional steps - as described in the following chapters.

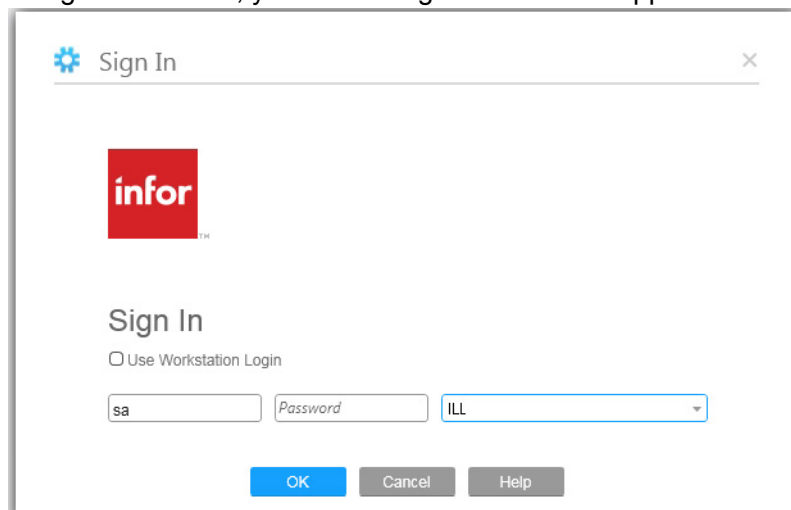
Users of each site need access to an end user client configuration for that site.

First Time Login

- 1 Using the administrative client on each utility server, start Mongoose. Use the shortcut on the desktop, or use the Windows menu:

Infor>Mongoose Client

- 2 Select the configuration for that site, and log in as **sa**. Enter the password defined for the sa user during Mongoose database server setup (page 12). This is the default password for the Mongoose sa user; you can change it later in the application.



Note: If you will be administering several site configurations from a single console, you may want to display the site configuration name in the window's title bar. This helps you keep track of which site you are currently working in. To set this up, select **View>User Preferences** from the Mongoose toolbar. In the Settings dialog, select **Show configuration name on caption**.

Specifying a Configuration Group for a Client

When users start Mongoose, they are presented with a list of configuration names to select from. If you set up configuration groups through the Configuration Manager utility, then you can specify a group name as part of the client connection information:

- If a client is using a WinStudio shortcut, you can specify the configuration group name using the **-gconfig-group** command line option. For example:
WinStudio.exe -gAcme
- If a client is connecting to the Configuration Server using HTTP, the configuration group may be specified as a query parameter in the URL. For example:
http://slutil/IDORequestService/ConfigServer.aspx?ConfigGroup=Acme
- If you are using XML to connect, you can specify the configuration group name as part of the GetConfigurations IDO request. For example:

```
<IDORequest>  
<RequestHeader Type="GetConfigurations" ConfigGroup="Acme"/>  
</IDORequest>
```

Specifying Other Startup Parameters for a WinStudio Client

If a client is connecting using a WinStudio shortcut, you can specify additional startup parameters as described in the "About WinStudio Startup Parameters" help topic.

Chapter 5: Setting Up Replication at Sites

5

During the Planning phase, you should have determined which type of data sharing you will use, and which categories of data will be replicated or shared.

Type of Replication	Page
Transactional Replication	32
Master site and shared _all tables (Follow the steps for transactional replication first)	36
Non-transactional replication	39

For detailed information about how replication works, see the *Replication Reference Guide*.

Setting Up Transactional Replication

Note: Skip this section if you are *not* using *transactional replication* (see the definition on page 7).

Log into *each site* as "sa" and perform the following steps. See the online help on the various forms for more information about any step.

Define Intranets

- 1 In the Intranets form, specify all intranets to be used in this multi-site system. The default intranet is DEMO. You may need additional intranets. During the Planning phase you should have determined the intranets you need and which sites will use them.

Note: Some forms, like this one, open "filter-in-place." Click the filter button in the toolbar to see all existing records and add new ones.

- 2 If you will be sharing *_all* tables and using a master site, define an intranet that will include all the sites that share tables. This cannot be an "External" intranet. All the sites on this intranet must use the same version of Mongoose.

Do not define the master site for the intranet yet - that will be done later.

- 3 In the **TaskMan Path** field for each intranet, specify the shared path to the folder containing TaskMan.exe on the utility server used by that intranet.

This path is usually set to `\\servername\Mongoose`.

For more information about how this field is used, see the *Multi-Site Planning Guide* information on TaskMan monitoring.

Any time you change the TaskMan Path, you should restart all TaskMan services that monitor configurations for application databases connected to this intranet.

Define Linking Between Sites

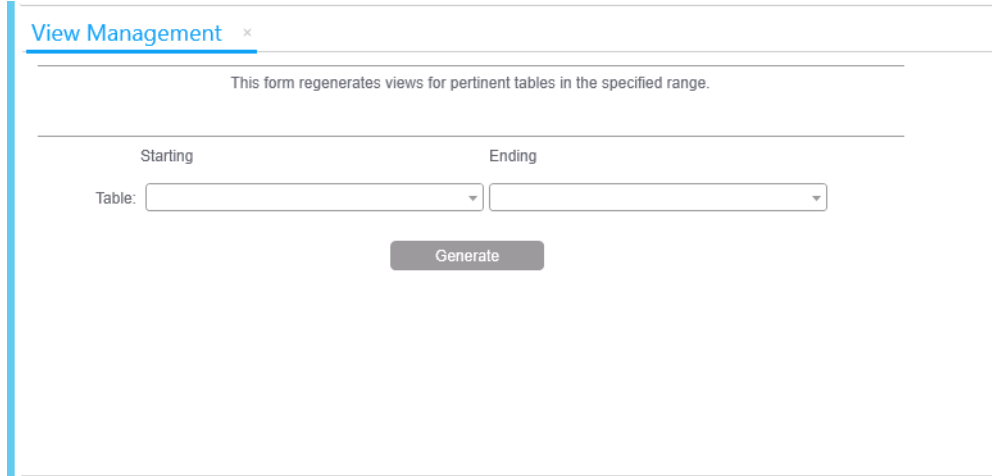
Use the Sites or Sites/Entities form to specify information about this site and the other sites that it communicates with:

- 1 On the Sites or Sites/Entities form, ensure that each site that was linked to the current site (see “Link Databases” on page 11) has a record in this form. This should happen automatically.
- 2 On the **System Info** tab, select the record for each site:
 - a Make sure the **Type**, **Time Zone**, **Database Name**, and **Forms Database Name** are set correctly.
 - b Set the **Intranet Name** to one of the intranets you defined on page 32.
 - c In the **Configuration Name** field, specify the name of a configuration that the system will use as the target configuration when building IDORequest XMLs during replication and AES event handling. If you leave this field blank, the system expects that a configuration exists with exactly the same name as the site name, and it uses that as the target configuration.
- 3 Review the list of linked sites shown in the **Link Info** tab. If you used the Infor Mongoose Configuration Wizard on the database server to set up linked sites (page 11), this tab shows those links, which are used in transactional replication between the currently selected site record's database and other site databases.
 - The *local site*'s record should show all the other databases linked to this site.
 - The records for *other* sites should show the link to the local site.
 - The **Linked Server Name** is the name of the *database server* where the site database resides.
 - On the *local site*'s record, you can add any other needed links between this local site and other site databases that will perform *transactional* replication with this database. If you later make changes to this record, the changes can be replicated to other sites through manual replication of the Site Admin replication category.

Regenerate Views

Use the **View Management** form to regenerate table views.

- 1 At one site in each application database, open the **View Management** form.

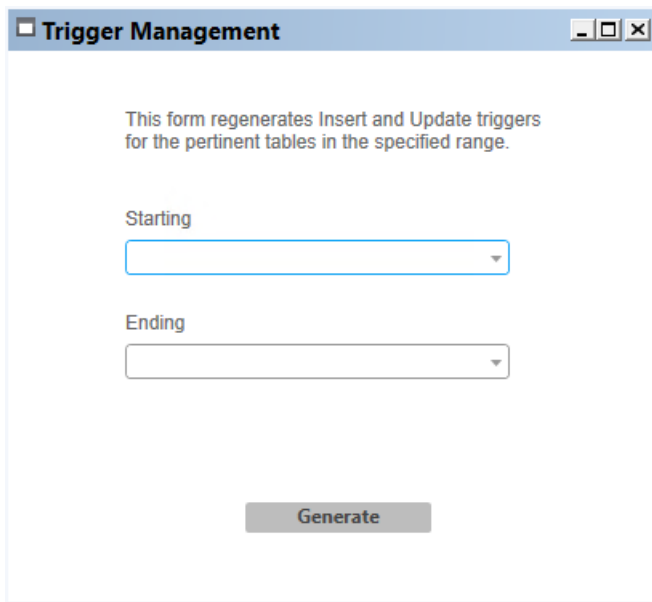
A screenshot of a web browser window showing the 'View Management' form. The browser's address bar displays 'View Management' with a close button. The form has a title bar 'View Management' and a subtitle 'This form regenerates views for pertinent tables in the specified range.' Below the subtitle, there are two dropdown menus labeled 'Starting' and 'Ending'. A 'Table:' label is positioned to the left of the 'Starting' dropdown. A 'Generate' button is located at the bottom center of the form.

- 2 Leave the range fields blank and click **Generate**.

Regenerate Triggers

Use the **Trigger Management** form to regenerate table triggers.

- 1 At one site in each application database, open the **Trigger Management** form.

A screenshot of a web browser window showing the 'Trigger Management' form. The browser's address bar displays 'Trigger Management' with a close button. The form has a title bar 'Trigger Management' and a subtitle 'This form regenerates Insert and Update triggers for the pertinent tables in the specified range.' Below the subtitle, there are two dropdown menus labeled 'Starting' and 'Ending'. A 'Generate' button is located at the bottom center of the form.

- 2 Leave the range fields blank and click **Generate**.

Set Up Replication Categories and Rules

- 1 Familiarize yourself with the categories listed in the Replication Categories form. Be aware of what categories are available and what each one handles. Add categories appropriate for your application.

Note: Adding new categories and rules requires significant knowledge of the database and business rules. When multiple tables are related, the order of replication can be critical. Improperly creating and executing replication categories and rules can result in catastrophic loss of data. Ensure that all such development and testing is performed in a controlled pilot environment prior to implementation. We strongly suggest that you contact Infor Consulting Services for assistance.

- 2 In the Replication Rules form, set up rules for the **Site Admin** category from this site to all other sites where replication will take place.

Add more rules from this site to other sites, as appropriate.

- For the **Interval Type**, select **Transactional**.
- You should have created a ReplicationRules spreadsheet as part of your Planning phase. You can save time by importing the appropriate rules for each site from the spreadsheet into the grid on the form at that site. See Appendix C, "Data Load To/From Spreadsheets," for more information.
- If a rule refers to a site that is not set up with matching information in the Sites or Sites/Entities form, the system will give a validation error when you try to save it.
- Even if a category contains _all tables that you plan to share, you generally still want to include replication rules for the category. The shared _all tables in the category will not be replicated; however, many categories contain additional base tables or stored procedures that are needed to perform certain functions. (See the example on page 38.)

- 3 Go to the Replication Management form and click **Regenerate Replication Triggers**.

Note: Finish this step for all sites before continuing with the next step.

Populate Parameter _All Records at Other Sites

In the Update _All Tables form at each site, repopulate certain _all tables.

Note: You must perform this step if the Sites or Sites/Entities form lists at least one site that is not in the same application database, even if you are not currently replicating any data to/from that site. The form lists only those tables that are shareable. If all of the internal sites (that is, sites on an intranet that is not set to "External") which are listed on the Sites or Sites/Entities form reside in the same application database, this should be an empty list.

In an initialized database, the parameters tables already have one record defined for the local site. Because the record already exists, it will not be created at other sites through normal replication. Repopulating the tables ensures that a parameter record for the local site exists at other sites where it is needed. Use these steps:

- 1 Click **Deselect All** to deselect all tables in the form.

- 2 Select the **Initialize _All Parameters** replication category and click **Select By Category** to select the parameters tables.
- 3 Make sure *nothing else* is selected (including Disable Replication or a Site name).
- 4 Click **Repopulate Tables** to repopulate each selected parameters _all table at the local site and to create or repopulate the local site record at any other sites to which the local site is replicating a category that contains the parameter table.

Back Up Databases

At this point, it may be wise to back up all of your databases, so you do not have to re-add all of your replication setup if problems occur later in the implementation.

Setting Up a Master Site and Shared _All Tables

Note: Skip this section if you are *not* using a *master site* or *intranet shared tables*.

See the online help on the various forms for more information about any step.

Set Up All Sites on the Intranet

At all sites on the intranet where sharing will take place (including the master site), perform all the steps under “Setting Up Transactional Replication” on page 32 through 36.

Note: If you already have a shared master site and intranet, and you just want to add a new site to it, see Appendix E, “Adding a Site or Entity to an Existing Shared Tables Intranet.”

Then return to this page for additional instructions.

Set Up the Master Site

Log in to the site that you want to make the master site where the shared tables will exist for the intranet. Then follow these steps at that site *only*:

- 1 Verify the information on the **Link Info** tab on the Sites or Sites/Entities form at the *master site*:
 - a On the Link Info tab for the *master site's record*, each of the other sites on its intranet should be listed as linked sites. For example, if the master site is ILL and the other sites on the intranet are CRP, USA, and CAL, the local site record for ILL would list all the other sites on its intranet.

- b On the Link Info tab for *other site records on the shared intranet*, the master site should be listed.

Sites on *other intranets* do not require Live Link information at the master site.

- 2 On the Intranets form, select this site's intranet. In the **Master Site** field, specify the master site where shared tables will reside.

The following must be true in order to add or change a value in this field:

- You must be logged into the site that you want to specify as the master site.
- This site must be defined as being within the selected intranet. Live links must be set up (on the Sites or Sites/Entities form) between this site and the other sites in the intranet.
- No table within the intranet is currently shared.

Note: After you define the master site for an intranet on the master site's Intranets form, the master site name displays, but cannot be changed, on the Intranets form at all other sites. (This assumes you have set up replication of the Site Admin category from the master site to the other sites in the intranet, or the sites are in the same application database.)

Set Up Shared _All Tables

On the Intranet Shared Tables form at the *master site*, select this site's intranet. A list of the _all tables that can be shared displays. (Not every _all table is listed; some are not available for sharing.)

Note: If you share _all tables or user tables, all sites in the same database must be on the same intranet.

- 1 For tables that you want to be shared between all sites on the intranet, select **Shared**.
Another way to choose the shared tables is to select the replication categories that you want to share. When you select a Replication Category from the drop-down list and click **Select by Category**, the system marks _all tables in that category as Shared.
- 2 When you have selected all the tables you want to share, select **Actions>Save**.
- 3 Click **Process**. This copies information from the tables at the other sites to the master site's table, deletes the table from the other sites (creating views into the master site tables instead), and regenerates the replication triggers for the other sites. If you have selected several tables and have many sites on this intranet, processing may take a while. The Processing Step area displays the system's progress.

Caution: During processing, the selected tables are removed from all sites on the intranet except the master site. Unsharing (rebuilding the tables at the using sites) is time-consuming - so be very sure that you have everything set the way you want it before clicking the Process button.

During processing, the system validates live link setup between the master site and the using sites of an intranet. If it finds a problem, an error message displays and nothing is processed; fix the link and then click **Process** again.

The Processed field indicates which tables have been processed - for example, if shared, they are now resident only in the master site's database. (Once a row on the form is marked as Processed, subsequent "Process" runs will not reprocess that row.)

After processing all the tables and sites, the system regenerates the replication triggers at the master site.

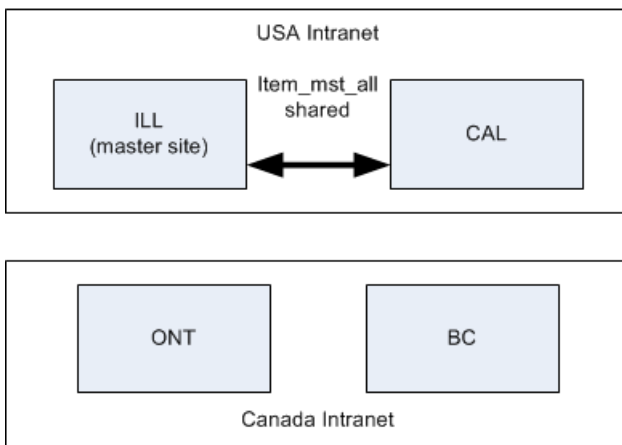
Set up Replication to Sites in Other Intranets

If there are other intranets with sites that want to replicate (not share) `_all` table data to/from sites in the sharing intranet:

- For tables that are shared, set up replication categories/rules between the *master site* and the sites on the other intranets. (You probably will also need to set up the same categories/rules between the specific shared site and the sites on other intranets; see the example below.)
- For tables that are not shared, set up replication categories/rules between any/all of the sites in the shared intranet and the sites on the other intranets.

Example

In the setup shown below, the following are true:



- If CAL needs visibility into BC's data, replication rules for a category containing the appropriate table should be set up from BC to ILL.
- If BC needs visibility into CAL's data, replication rules for a category containing the appropriate `_all` table should be set up from ILL to BC.
- If BC and CAL interact in other ways, you need additional replication rules between CAL and BC, since additional tables and stored procedures are needed to perform these functions.
- If ONT needs visibility into CAL's data which is not in a shared table, replication rules for a category containing the appropriate `_all` table should be set up from CAL to ONT.

Setting Up Non-Transactional Replication

Skip this section if you are *not* using *non-transactional replication* (see the definition on page 7).

If you are using non-transactional replication only to interface to another Infor product, refer to the setup steps in the appropriate product's implementation guide, instead of the steps below.

Log into *each site* as "sa" and perform the following steps. See the online help on the various forms for more information about any step.

Set Up a Replication User

Note: Before completing this step, you will need to install a license. See Chapter 6, "Setting Up Licensing at Sites," for instructions.

Create a user account that will be used during replication between the sites. Assign this user the appropriate permissions to allow replication data to complete the transaction. This protects the local site from another site inappropriately adding or modifying data on the local site without permission.

- 1 Add a replication user called **repl_user** on the Users form at *all sites where non-transactional replication will be used*. Initially, make that user a super user and do not assign any authorization groups.

If you have multiple administrators for different sites, you can later decide to revise the permissions and groups assigned to this replication user, in order to limit the data that certain sites are allowed to send to other sites. However, until everything is set up and working properly, leave this user as a super user who can access everything.

- 2 Click **User Modules** to specify the license module this user is assigned to.

Generally, this should be **MGCoreAutomation**

Set Up System Types

If this site will be replicating data to other applications, or to other versions of Infor Mongoose, use the System Types form to create a system type for each other application/version. This system type is used when defining the target site for the application in the Sites or Sites/Entities form. You should also define a system type for the Mongoose application. (The source and target system types are used in the names of XSL stylesheets that can be used to transform the XML data; see the System Types help topic for more information.)

The list of system types should match at all sites. System Types are replicated by the Site Admin category rules, so create a minimum list now and add more types later as needed.

Define Intranets

- 1 In the Intranets form, specify all intranets to be used in this multi-site system. The default intranet is DEMO. You may need additional intranets. During the Planning phase you should have determined the intranets you need and which sites will use them.
- 2 Specify connection information used in data requests to and from the other sites:
 - **External:** Select this field if the intranet is an external intranet, meaning that it allows applications other than those based on the Mongoose (WinStudio) toolset. Replication done on an external intranet must be non-transactional, and an external intranet cannot have a master site.
 - **Transport:** This field, which is enabled if you selected External, allows you to select either HTTP or ESB as the transport protocol for exchanging data with external applications.
 - HTTP POST is used to handle most non-transactional replication. If you select HTTP, you then define a set of MSMQ instances, MSMQ listeners, and URLs for the intranet.
 - ESB is used for connecting to messaging middleware through Infor ION.
 - HTTP: Options in this group are enabled only if you chose HTTP as the Transport method:
 - **Queue Server:** Enter the name of the MSMQ server for the intranet. If left blank, this field defaults to the utility server running this application, and the private queue area. If you have multiple intranets on different servers, include the utility server machine name in the path (for example, *machinename\PRIVATE\$*).
 - **Private Queue:** We recommend that you select this field, since queues in non-transactional replication are used only by Mongoose services on the same machine. Setting up public MSMQ instances requires further ActiveDirectory setup, which is not required by the Mongoose services.
 - **URL:** This is the web address to which the XML request/response documents will be posted. When replication occurs between sites on different intranets, XML documents are routed to this URL for the target site. It may also be used by system integrators for programmatic access to a site on this intranet.

This URL could be an active server page (ASP) on the external system that receives and processes the XML from the Mongoose message queue. The processing done at this URL is up to you; for example, an ASP could map the data into the proper format for the external system, or it could write the XML documents to a location on its server for later processing.

In many cases, this field provides the address of the default InboundQueue ASP in the virtual directory that was installed with the web server components.
 - **Direct IDO URL:** This is the web address of the synchronous XML interface to this Intranet. It is used primarily by WinStudio clients which are configured to connect via the Internet. It may also be used by system integrators for programmatic access to a site on this intranet.

You must specify a fully qualified URL. It should be the address of the ConfigServer, which is deployed on your utility server when you include the web server feature of the utility server setup. For example, if the machine name is UTILSERVER, then the URL might look like this:

`http://UTILSERVER/IDORequestService/ConfigServer.aspx`

Or if you want to use secured http - a good idea if this server will be open to the internet for http traffic - use this format:

`https://UTILSERVER/IDORquestService/ConfigServer.aspx`

- **Master Site:** See “Set Up the Master Site” on page 36.
 - **Tenant ID:** This field is used during integration to other applications through business object documents (BODs). See the online help for more information.
- 3 Specify the **TaskMan Path** for each intranet. This is the shared path to the TaskMan folder on the utility server used by that intranet.

This path is usually set to `\\servername\Mongoose`.

If you change the value in this field, restart all TaskMan services that monitor configurations for application databases connected to this intranet.

Define Linking Between Sites

Use the Sites or Sites/Entities form to define sites in your system.

- 1 In all records for sites that will be participating in non-transactional replication, specify a system type. On the **System Info** tab, **System Type** field, select from the list you created in the System Types form. The system type you choose should exist at all of your sites.

If this is a new site record being added for non-transactional replication, specify additional information about the site, as applicable, on this tab.

In the **Configuration Name** field, specify the name of a configuration that the system will use as the target configuration when building IDORquest XMLs during replication and AES event handling. If you leave this field blank, the system expects that a configuration exists with exactly the same name as the site name, and it uses that as the target configuration.

- 2 On the **Site User Map** tab, specify the Mongoose user name under which the *local site* (the one you are currently logged into) communicates with the *target site* (the site whose record is currently displayed). **From Site** is the local site. **User Name** is the local user that the local site specifies when using the IDO connection to perform the requested transaction on the target site. The user name and the password for that user must be the same on both the local and target sites. For now, specify the user name `repl_user` (set up on page 39) for all sites.

- If you use the preferred “list of configurations” method, where the system uses information from the database into which users are logged, ensure that the local site has Site User Map rows associated with each of the “From Sites” from which the local site will receive *non-transactional* data. This scenario is illustrated by this example:

You have three databases:

- DB1 has one site, Shanghai.
- DB2 has one site: Windsor.
- DB3 has two sites, CMH and CIN.

Communication between the sites is set up like this:

- CMH and CIN communicate through synchronous (transactional) replication with each other.
- CMH and CIN communicate through asynchronous (non-transactional) replication with Shanghai and Windsor.
- Shanghai and Windsor communicate through asynchronous (non-transactional) replication with CMH and CIN and with each other.

Each database contains five site records:

- Shanghai
- Windsor
- CMH
- CIN
- INFORBUS (used for communications with ION)

The Sites or Sites/Entities form includes these Site User Map rows for each database:

DB1 site record	Contains these Site User Map rows
Shanghai	From Site: Windsor From Site: CMH From Site: CIN
Windsor	
CMH	
CIN	
INFORBUS	From Site: Shanghai From Site: Windsor From Site: CMH From Site: CIN
DB2 site record	Contains these Site User Map rows
Shanghai	
Windsor	From Site: Shanghai From Site: CMH From Site: CIN
CMH	
CIN	
INFORBUS	From Site: Shanghai From Site: Windsor From Site: CMH From Site: CIN
DB3 site record	Contains these Site User Map rows
Shanghai	
Windsor	
CMH	From Site: Shanghai From Site: Windsor
CIN	From Site: Shanghai From Site: Windsor

DB3 site record

INFORBUS

Contains these Site User Map rows

From Site: Shanghai

From Site: Windsor

From Site: CMH

From Site: CIN

Update Views

Use the **View Management** form to regenerate table views.

- 1 At one site in each application database, open the **View Management** form.

View Management ×

This form regenerates views for pertinent tables in the specified range.

Starting Ending

Table:

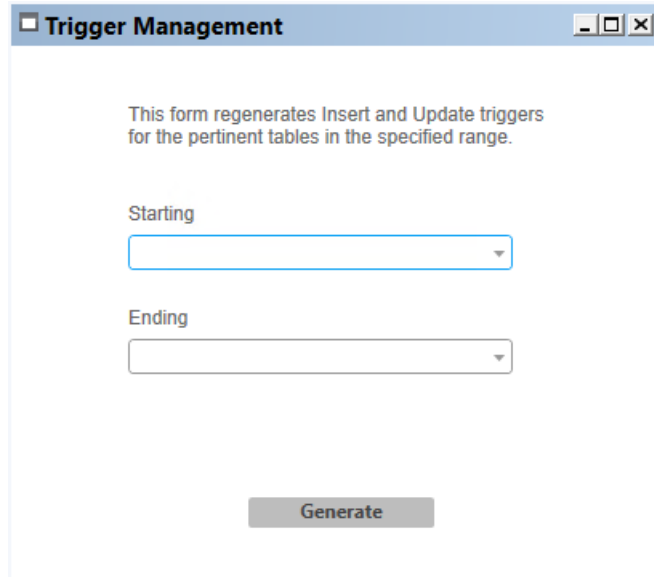
Generate

- 2 Leave the range fields blank and click **Generate**.

Regenerate Triggers

Use the **Trigger Management** form to regenerate table triggers.

- 1 At one site in each application database, open the **Trigger Management** form.



- 2 Leave the range fields blank and click **Generate**.

Set Up Replication Categories and Rules

- 1 Familiarize yourself with the categories listed in the Replication Categories form. Be aware of what categories are available and what each one handles. Add categories appropriate for your application.

Note: Adding new categories and rules requires significant knowledge of the database and business rules. When multiple tables are related, the order of replication can be critical. Improperly creating and executing replication categories and rules can result in catastrophic loss of data. Ensure that all such development and testing is performed in a controlled pilot environment prior to implementation. We strongly suggest that you contact Infor Consulting Services for assistance.

- 2 In the Replication Rules form at each site, set up non-transactional rules for the **Site Admin** category between this site and all other sites where non-transactional replication will take place.

Add more non-transactional rules from this site to other sites, as appropriate.

- For the **Interval Type**, select anything other than Transactional, and specify the appropriate interval and start date/time.
- You should have created a ReplicationRules spreadsheet as part of your Planning phase. You can save time by importing the appropriate rules for each site from the spreadsheet into the grid on the form at that site. See Appendix C, "Data Load To/From Spreadsheets," for more information.

- If a rule refers to a site that is not set up with matching information in the Sites or Sites/Entities form, the system will give a validation error when you try to save it.
- 3 At each site, go to the Replication Management form and click **Regenerate Replication Triggers**.

Note: Finish this step for all sites before continuing with the next step.

Populate Parameter _All Records at Other Sites

In the Update _All Tables form at each site, repopulate certain _all tables.

Note: You must perform this step if the Sites or Sites/Entities form lists at least one site that is not in the same application database, even if you are not currently replicating any data to/from that site. The form lists only those tables that are shareable. If all of the internal sites (that is, sites on an intranet that is not set to "External") which are listed on the Sites or Sites/Entities form reside in the same application database, this should be an empty list.

In an initialized database, the parameters tables already have one record defined for the local site. Because the record already exists, it will not be created at other sites through normal replication. Repopulating the tables ensures that a parameter record for the local site exists at other sites where it is needed.

- 1 Click **Deselect All** to deselect all tables in the form.
- 2 Select the **Initialize _All Parameters** replication category and click **Select By Category** to select the parameters tables.
- 3 Make sure nothing else is selected (including Disable Replication or a Site name).
- 4 Click **Repopulate Tables**. This repopulates each selected parameters _all table at the local site and creates/repopulates the local site record at any other sites to which the local site is replicating a category containing the parameter table.

Note: At this point, back up all databases to save your replication setup in case problems occur later in the implementation.

You should have requested and received the proper licenses for each of your sites from your Infor representative. Use the Licensing spreadsheet from your Planning phase to help with this.

Intranet Licensing

If you do not plan to use intranet licensing, skip to “Site Licensing” on page 48.

Note: Intranet licensing is required if you plan to use the Intranet Shared User Tables feature.

Prerequisites

If you plan to use the Intranet Licensing feature, the following must be true. Instructions for all of these prerequisites are found in Chapter 5, “Setting Up Replication at Sites:”

- You must have set up an intranet with a master site.
- All sites that will use the intranet licensing must be set up (in their Sites or Sites/Entities form) to use that intranet.
- All sites on that intranet must be replicating the Site Admin category, in order to recognize the master site.
- All sites that will use the intranet licensing must specify an application database name in their Sites or Sites/Entities form.

Turn on Intranet Licensing at the Master Site

- 1 At the *master site only*, log in and follow the steps under “Site Licensing” on page 48 to apply a license.
- 2 In the *master site*’s Sites or Sites/Entities form, select the **Intranet Licensing** check box on the local site record and save the record.

Turn on Intranet Licensing at Other Sites

- 1 Make sure all other users are logged out of the sites.
- 2 Log into any other *sites* where you want to use the master site's licensing. On the Sites or Sites/Entities form at each site, select the *local site* record and select **Intranet Licensing**. Save the record.
- 3 Log out of all the sites, including the master site, and then log back in. This picks up the changes to certain licensing records, which are now stored only on the master site. The licensing forms are now disabled in the non-master sites on the intranet.

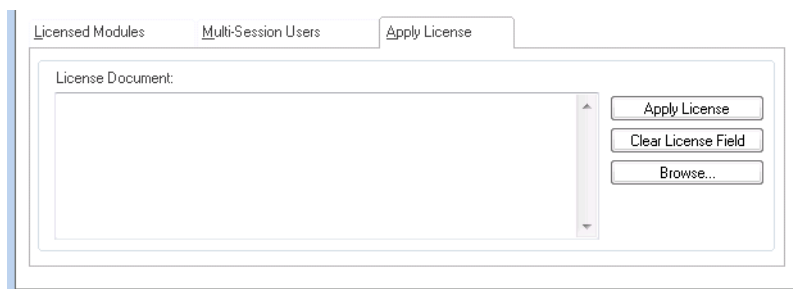
Site Licensing

Note: Licensing is applied to an application database.

Initial Steps

Perform the following steps *at each site*. Or, if you are using intranet licensing, perform these steps only at the master site.

- 1 Go to the License Management form, where you will apply the appropriate licenses.
- 2 Select the **Apply License** tab.

The screenshot shows a web application interface with three tabs: 'Licensed Modules', 'Multi-Session Users', and 'Apply License'. The 'Apply License' tab is active. Below the tabs is a large text area labeled 'License Document:'. To the right of this text area are three buttons: 'Apply License', 'Clear License Field', and 'Browse...'. The 'Apply License' button is highlighted.

- 3 Paste your license document into the License Document field. You can cut and paste the information from the license document into this field, or you can click **Browse** and select the document from your drive. The document is encrypted and every character counts, so if you copy and paste, make sure to copy the entire contents of the document.
- 4 Click **Apply License**. Any previous licensing that you had for this site is deleted and replaced with the licensing defined in the new license document.

The **Multi-Session Users** tab lists any multi-session users defined by your license.

For information about this user and about how concurrent or named user licensing works, see the *Infor Mongoose Licensing* document.

Additional Steps

- 1 At each site, use the **Optional Modules** form to enable the licensed modules for that site.

Optional Module Name	Enabled	System Record	Licensed	Module Description
1 SyteLineCCI	☐	☒	☒	
2 SyteLineCN	☐	☒	☒	
3 SyteLineFSP	☐	☒	☒	
4 SyteLineJP	☒	☒	☒	
5 SyteLineMX	☐	☒	☒	
6 SyteLinePP	☐	☒	☒	
7 SyteLineTax	☐	☒	☒	
8 SyteLineTH	☒	☒	☒	
9 SyteLineWB	☒	☒	☒	

- 2 If there were active WinStudio sessions when you applied the license, perform the following steps *on the utility server*. If there were not any active sessions, you can skip these steps.
 - a Open the Configuration Manager utility (**Start>All Programs>Infor>Tools>Configuration Manager**).
 - b Select the **Utilities** tab.

Configuration Manager

Configurations Applications Web Servers Configuration Groups **Utilities**

☒ All Configurations
☐ Configuration
☐ Configuration Group

☐ IDO Runtime on local machine

Discard IDO Cache

Publish Form Change

Help

Path: C:\Documents and Settings\All Users\Application Data\Mongoose

- c Discard the Runtime Service Cache for all configurations that reference the application database to which the new license has just been applied. To do this, select **All**

Configurations and **IDO Runtime on local machine**, and then click **Discard IDO Cache**. See the Configuration Manager online help for more information about the options on this tab.

View Sites Where Users Are Connected

The License Modules per User Connection form has a Sites column that lets you monitor the site to which each user is connected:

License Modules per User Connection

Module User Count: _____

Module Name: Transactional

Module User Count: 53

License Modules Per User Connection

	User Name	Connection Date	Module Name	Site
1	bcombs	9/28/2016 12:00:51 PM	Transactional	DALS
2	pcoate	9/19/2016 1:37:45 PM	Transactional	DALS
3	pcoate	9/28/2016 11:10:06 AM	Transactional	DALS
4	pcoate	9/27/2016 12:09:30 PM	Transactional	DALS
5	mzhang2	9/28/2016 5:29:45 AM	Transactional	DALS
6	mzhang2	9/27/2016 7:39:32 AM	Transactional	DALS

Session Management Concurrent User Summary

If Intranet Licensing is enabled, this form shows the site from which the users are connected. In non-intranet licensed environments, only connections from the current site are shown.

Chapter 7: Setting Up Users and Authorizations

7

If you plan to use a master site to maintain users for all sites on an intranet, set up the shared user tables as described below and then perform the other steps in this chapter at the *master site* only. If you are using Intranet Licensing, the users at all sites must also be defined at the master site. Then the users must be assigned to license modules on the master site (through the User Modules form).

If you are not using a master site, skip the first step and perform the other steps in this chapter at each site to set up users.

Setting Up Shared User Tables

Note: Intranet Licensing must be set up at the master site and all participating sites before you share user tables for the sites. Also, if you share user tables, all sites in the same database must be on the same intranet.

You can maintain user and group names and permissions, user-group mappings, user license module assignments, and application-specific user information at the master site of an intranet. To do this, you must share the user tables from all sites on the intranet; that is, set up views to the tables, which only reside at the master site:

- 1 Log in to the master site for the intranet where you want to share user tables.

2 On the Intranet Shared User Tables form, select the master site's intranet.

Intranet Shared User Tables

Intranet: SHARED

Master Site: ILL ILL Site

Shareable User Maintenance Tables

Table Name	Shared	Processed	UserId Column	GroupId Column
1 AccountAuthorizations_mst	☒	☐	Id	Id
2 GroupNames	☒	☐		GroupId
3 user_local_mst	☒	☐	UserId	

Non-Shareable Tables Containing References to UserId or GroupId

Table Name	Update Referenced Id	Processed	UserId Column	GroupId Column
1 adp_parm_mst	☒	☐	UserId	
2 cad_parm_mst	☒	☐	UserId	
3 ctc_log_mst	☒	☐	UserId	

Process Options

☒ Set up shared user tables Status: User tables are not shared

Processing Step:

Two lists of tables display:

- The top grid lists user tables that can be shared. You can clear the Shared option for some tables, but not all of them, as described below.
 - The bottom grid lists tables that contain a column whose base domain is UserNames.UserId or GroupNames.GroupId. The bottom grid is used during setup of shared user tables to identify the tables and columns that may need to be updated if records that were formerly defined in the Usernames or Groupnames tables in the non-master site are moved to the master site, but with different UserId or GroupId values.
- 3 If you have custom tables that contain a column whose value comes from base domain UserNames.UserId or GroupNames.GroupId, add your custom table and its associated ID column to the Non-Shareable Tables grid. We recommend that your custom tables refer to the Username or Groupname columns, rather than the ID columns, because the distinct list of Usernames and Groupnames across intranet sites is always the same, whether they are stored in shared tables or per site, and therefore no changes are required for data referencing this base domain.
 - 4 Select **Set up shared user tables**, which automatically selects **Shared** for all tables in the top grid and **Update Referenced ID** for all tables in the bottom grid.
 - 5 In the top grid, clear the Shared check box for any tables that you do not want to share between sites.

Note: Be aware of how this works. For the Usernames, GroupNames, UserEmail, UserModules, UserPasswordHistory, UserCalendar, and UserTask tables:

- If you do not share these tables on the Intranet Shared User Tables form, these tables are:
 - Not shared across multiple sites, if one site per database
- If you do share these tables on the Intranet Shared User Tables form, these tables are:
 - Shared across multiple sites, if one site per database

For the AccountAuthorizations_mst, user_local_mst, and UserGroupMap_mst tables:

- If you do not share these tables on the Intranet Shared User Tables form, these tables are:
 - Not shared across multiple sites, if one site per database
- If you do share these tables on the Intranet Shared User Tables form, then you can optionally share each of these tables across multiple sites.

- 6 Click **Process** to copy information from the tables at the other sites to the master site's table, delete the table from the other sites (creating views into the master site tables instead), and regenerate the replication triggers for the other sites. (Removing tables and setting up views does not happen if the non-master site is in the same application database as the master site.)

If you have many sites on this intranet or many users and groups, processing can take a while to complete. The Processing Step area displays the system's progress.

During processing, the system validates link setup between the master site and the using sites of an intranet. If it finds a problem, an error message displays and nothing is processed; fix the link and then click **Process** again.

The **Processed** field indicates which tables have been processed - for example, if shared, they are now resident only in the master site's database. (After a row on the form is marked as Processed, subsequent "Process" runs do not reprocess that row.)

The **Status** field indicates that the user tables are either shared or not shared.

After processing all the tables and sites, the system regenerates the replication triggers at the master site.

- 7 After processing is complete, you must reapply a valid license document on the master site.
- 8 In the *master site*, open the View Management form and regenerate views at the master site, so that the views reflect the changes to table sharing.
- 9 In the *master site*, open the Trigger Management form and regenerate table triggers at the master site.

Set Up Password Parameters

Use the Password Parameters form to specify password expiration and complexity rules for all users. Be sure to explain the rules you choose to the users of your system.

The password parameter data is replicated as part of the Site Admin category.

Add Common Users

- 1 In the Users form, set up other common users - those who will be working in multiple sites. You can also set up local users now, if you want.

These users are in addition to the sa user (and the repl_user, if set up in Chapter 5, "Setting Up Replication at Sites"), which have full access to the local site.

If the user_local_mst table is not shared in the Intranet Shared User Tables form, then you must update the multi-site group and other user fields at each site where the user record exists, even for sites in the same database. If the AccountAuthorizations and/or UserGroupMap tables are not shared, then you must update authorizations and groups at each site.

- You must specify a default authorization group for each user.
 - Define authorization groups on the **Groups** form. After you define a group, select it and click the **Group Authorizations** button to specify the forms that members of this group can access.
 - If you do not set up at least one authorization group for a user, then the user has access to no forms, unless the user is set up with super user access.
- For developer users, if you are using a source control system, specify the user's login information for the source control system.
- You cannot replicate the usernames or groupnames tables because those tables use identity columns. However, to simplify the initial setup of multi-site users, you could add the multi-site users and groups in one site, export the data to Excel spreadsheets, and import into sites in other databases. (See Appendix C, "Data Load To/From Spreadsheets," for more information.)

For sites in separate application databases, the user password, email addresses, and workstation login can be set up in one site and replicated to the rest

However, the user record, assigned user module, authorization group and other required fields must already be set up at the individual sites.

- 2 From the Users form, click **User Authorizations** to display the Object Authorizations for User form, where you set up object authorizations for each new user record. Settings made here override any group authorizations for this user.

Some of your users may require special access to certain forms that is not covered in the default group authorizations.

- 3 If you decided during the Planning phase to use *named user licensing*, return to the Users form and select a user record. Click **User Modules** to go to the User Modules form where you can set up licensing for the selected user.

Repeat this step for each user in the site. Use the Licensing spreadsheet from your Planning phase to help with this.

If you are using Intranet Licensing, the User Modules form is available only at the master site and not at the participating sites.

See the *Infor Mongoose Licensing Guide* for more information about how licensing works.

- 4 Run the User Authorization Report form if you want to have a list of authorization information for each user, for auditing purposes.

Setting up Users to Work with Infor Ming.le

See the *Infor Mongoose Configuration Guide for Infor Xi Platform* for information about using BODs to transfer user information between Mongoose and Infor Ming.le.

Chapter 8: Setting Up Other Modules and Products

8

Once you have your basic Mongoose multi-site system set up, you can install and configure other modules and products according to the documentation listed below. Note that this may not be a complete list of all available modules and products.

To set up this	Refer to this information
Infor Xi Platform (Infor Ming.le, ION, Infor Document Management, In-Context BI)	<i>Infor Mongoose Configuration Guide for Xi Platform</i> <i>Infor Mongoose Content User Guide for Xi Platform</i>
Application Event System	<i>Application Event System Guide</i> , online help
Infor Framework FAX Service	<i>Infor Mongoose Installation Guide</i>
.NET Web Service	<i>Infor Mongoose Installation Guide</i>
Online help	<i>Infor Mongoose Installation Guide</i>

Handling Replication Errors

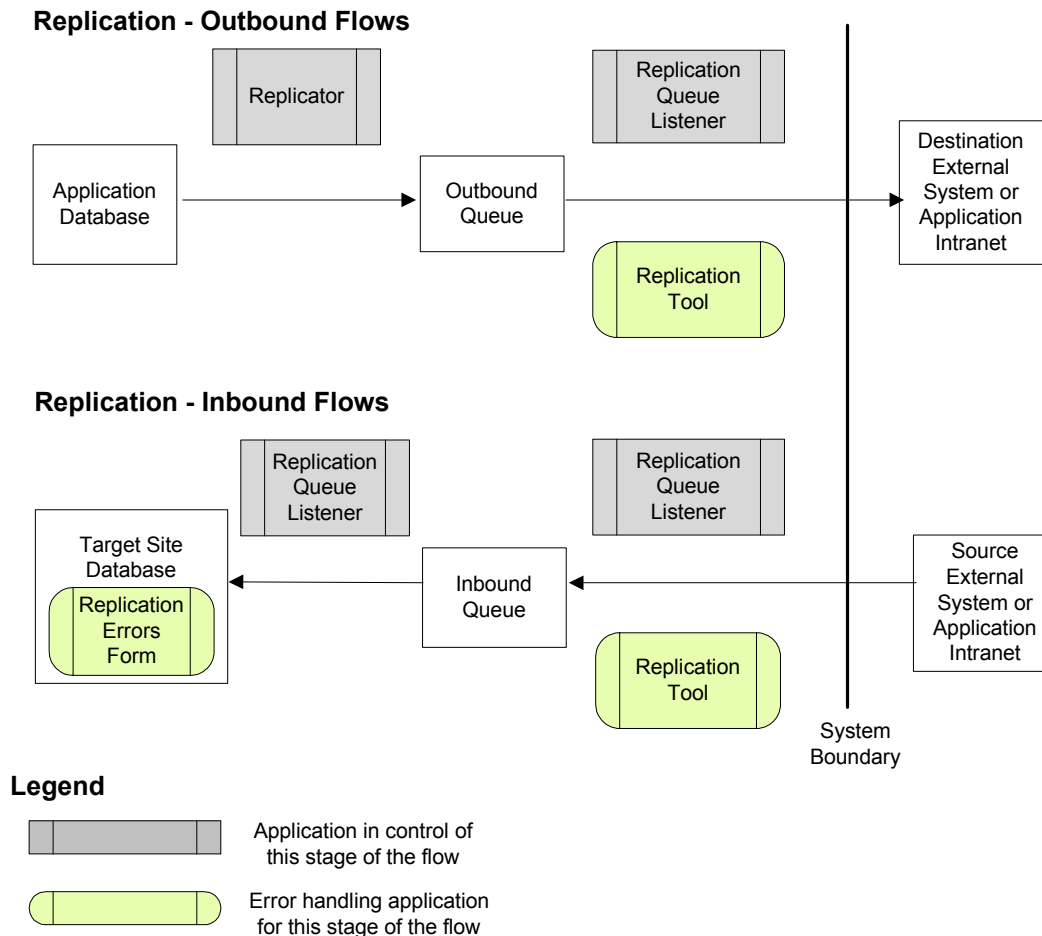
Errors During Transactional Replication

Validation errors relating to transactional replication generally display during data entry or saving of transaction records at the local site, and do not allow the user to continue.

However, if transactional replication appears to not be working (that is, the data is not being moved to the target site as expected) and the cause is not immediately apparent, use the “Replication Troubleshooting Checklist” on page 60 to help find the problem.

Errors During Non-Transactional Replication

If errors occur when sending or receiving XML request documents through Mongoose's replication system, the following processes handle the errors.



Replication Troubleshooting Checklist

Problem: *Replication does not seem to be working.*

Solution: View the appropriate _all table(s) using SQL Server tools, to determine if data is being populated.

If replication is not working, the table typically will not contain the newly entered data records. In this case, there probably is a problem with the replication rules.

For replication problems, check the following areas:

1. Configuration Manager utility (page 16): Verify, for each site and entity, that either 1) a default configuration name is defined for that site on the Sites/Entities form, or 2) you have created at

least one configuration name that exactly matches the site/entity ID, including case, and that uses the site/entity's application database.

2. Replication Rules form (page 35 and 44): Make sure the rules are set up properly between the site where the data was added and the one where it was supposed to appear.

Are the proper categories being replicated?

- a. View the Replication Rules form in the source site (for example, if you're expecting to see CAL data in ONT, view the CAL database), and determine if there is a rule set up with Target Site equal to the site where you expect to see the data, and with Category equal to the category as determined above.
- b. If such a record *does not exist*, this means the behavior was expected. You can create a new rule if desired, and retest to verify that the data was passed over.
- c. If the record *exists*, check its **Interval Type**:
 - If the Interval Type is **Transactional**, there may be another problem with the replication setup. Continue through the other steps in this check list.
 - If the Interval Type is **Immediate**, there may be a problem with the setup of the replication services.
 - If the Interval Type is something other than Transactional or Immediate, wait until the time specified in **Start Interval At** before testing again. If the interval time has passed and the change has still not occurred on the target machine, there may be a problem with the setup of the replication services.
3. Sites or Sites/Entities form (page 35 and 44): Make sure the database name and the linked servers (for transactional replication) or user maps (for non-transactional replication) are set up correctly.
4. Replication Management form (page 35): You *must* click the **Regenerate Replication Triggers** button on this form after any replication rules have been changed. Failure to do so will result in replication not working properly.
5. Find Replication Setup Issues form (page 67): Use this form to find problems relating to the setup of replication from the current site to another site for a particular database table or stored procedure.

Not all problems found by this utility are valid in all situations. For example, it checks to see if a user name is set up between the sites, which applies only if non-transactional replication is being used.
6. Replication Errors form (page 66): This form shows the data that failed to submit to a target site during non-transactional replication. Use the form to fix and resubmit data, or delete it.
7. Replication Tool (page 65): This tool on the utility server is used to troubleshoot non-transactional replication.
8. ReplicatedRows3 and ShadowValues tables: When something is to be replicated through *non-transactional* replication, it gets copied to the ReplicatedRows3 and ShadowValues tables. See the *Replication Reference Guide* for details. If the data is not there, it is likely that either your replication rules aren't set properly or that you need to click the Regenerate Replication Triggers button on the Replication Management form.

9. Services: If you are using *non-transactional* replication, make sure the Infor Replication Queue Listener and Infor Replicator services are started on the utility server.
10. MSMQ or Message Queuing: Turn on journaling.
11. Event Viewer: Check the event viewer in the Windows Administrative Tools on the utility server or database server. The events that are logged may give you a clue about where replication failed.
12. UETs: If you are using user extended tables, and the UET information is not replicating properly, see the information about UETs and `_all` tables in the *Replication Reference Guide*.

Common Replication Solutions

Problem: *System performance is slow and you suspect replication may be causing this.*

Solution: Check the following:

1. Use the stored procedure `sp_spaceused` (in the Master database) to determine which tables consume a lot of space.
2. If there are `_all` tables (that is, replicated data) that contain very large numbers of records, then make sure you are not replicating unnecessary data.

Also look at the descriptions of how the tables in each category are used. If your large `_all` tables are used only for reports that you rarely or never require, you may be able to remove that object from the category.

Note: Before making changes to replication categories, seriously consider requesting help from your Infor MongOOSE provider or Infor Consulting Services.

Problem: *Replication failure due to SQL Linked Server error.* If your system returns an error like one of these, the local server probably no longer has a link to itself:

- Server '*servername*' is not configured for Data Access. Distributed Transaction Completed. Either enlist this session in a new transaction or the NULL transaction.
- New transaction cannot enlist in the specified transaction coordinator.
- The operation could not be performed because the OLE provider 'SQLOLEDB' was unable to begin a distributed transaction.

Solution: To fix this problem, run the following SQL commands on the App database:

```
exec sp_addserver servername, LOCAL
-- restart SQL Server.
EXEC sp_serveroption 'servername', 'Data Access', 'True'
```

Replace *servername* with the name of the local server.

Message: *Invalid Column name UF_xxxx.* If your system returns an error message similar to this during replication, you may have user extended tables (UETs) set up in one site but not in another.

Solution: You must have identical schemas in all sites to replicate data. If your system includes User Extended Tables (UETs), consider the following when replicating data to other sites.

If one site contains UET modifications to a table, but the same table in another site does not contain those modifications, errors probably will be generated during replication of that table's data from one site to the other.

For that reason, before you replicate data between sites, you should copy any UET schema changes to all sites involved in the replication. A stored procedure, ExportUETClassSp, assists in handling this process.

Problem: *Outbound BODs are not being sent from Infor Mongoose to another system.*

Solution: Follow these steps to determine the cause:

1. Does the BOD appear in the Replication Document Outbox?

(BODs are saved in the outbox after processing only if the BOD service is configured to not delete them.)

If Yes, continue to the next step.

If No, then do this:

- a. If this is a multi-site system, verify that you are checking the outbox on a site that is defined in the **Replicator/Inbound Bus Configurations** area in the Service Configuration Manager utility.
- b. From the Mongoose utility server, run the LogMonitor utility.
- c. In the Mongoose application, perform the processing that should send the BOD. (See the Replication Document Outbound Cross References form for a list of the application events that generate the BOD.)
- d. In the LogMonitor utility, look for ReplQLListener log information that might indicate an error in the BOD processing.

2. Is the Processed check box selected?

If Yes, the BOD was picked up by the BOD service. Check in that application(s) for details.

If No, then there is a problem that is preventing the BOD service from picking up the message; either the service is not running or the configuration and deployment of the sending/receiving applications is not set up properly in the receiving application(s).

Problem: *Inbound BODs are not being received or processed by Infor Mongoose.*

Solution: Follow these steps to determine the cause:

1. Does the BOD appear in the Replication Document Inbox?

(If there are many documents in the inbox, it can be difficult to find the BOD because currently you cannot filter by BOD name.)

If Yes, continue to the next step.

If No, then something prevented it from being delivered by the BOD service:

- a. If this is a multi-site system, verify that you are looking at the inbox on a site that is defined in the **Replicator/Inbound Bus Configurations** area in the Service Configuration Manager.
 - b. Check the BOD service logs to find failed message transmissions or information about why the failure occurred.
 - c. Check whether the BOD service is running.
 - d. Check whether the configuration and deployment of the sending/receiving applications is set up properly. See the appropriate integration guide for more information.
2. Is the Processed check box selected?
- If Yes, and the expected information does not appear in the appropriate Mongoose form or table, then there was an error in processing. Check the Replication Document Outbox for a Confirmation BOD. A confirmation BOD indicates an error. Check the following areas to determine the cause of the error:
- Windows application event viewer
 - LogMonitor utility on the utility server, if it is running
- If No, then either the Infor Framework Inbound Bus Service is not running, or it is still working through previous messages.

Problem: After adding a site in a different database, replication is not working

If all sites were previously in the same application database, and you add a site in a different application database, you must regenerate views and replication triggers, and repopulate _all tables.

Solution: Follow these steps in one site in each application database:

1. In the Trigger Management form, leave the fields blank and click **Generate** to regenerate table triggers. (This is not related to replication, but it must be done after adding a new site in another database.)
2. If any new replication rules were set up to/from the new site, open the Replication Management form and click **Regenerate Replication Triggers**.
3. In the View Management form, leave the fields blank and click **Generate** to regenerate views over tables.
4. Restart the replication services on the utility server.
5. In the Update _All Tables form, select all tables, set **Site** to the current site and click **Repopulate**.

Note: This step must be performed at every site and entity.

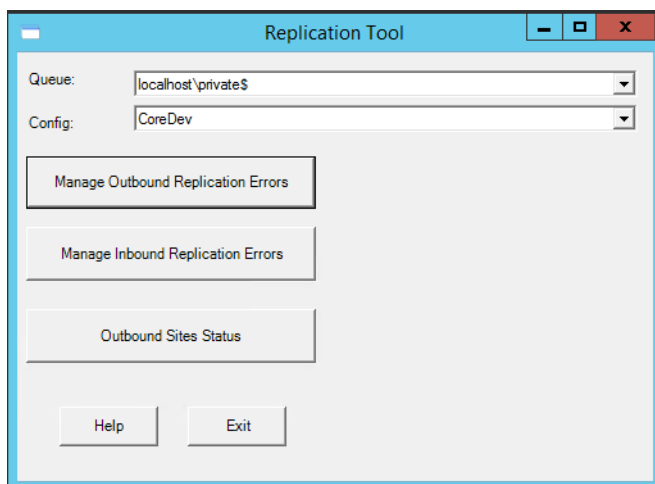
Forms and Utilities Used to Troubleshoot Replication

Replication Tool (Inbound or Outbound Flow)

If errors happen while Mongoose replication is processing an inbound or outbound request, you can use the Replication Tool on the utility server to:

- View, correct, and resubmit inbound and outbound XML request documents
- View the status of sites linked to this site for replication

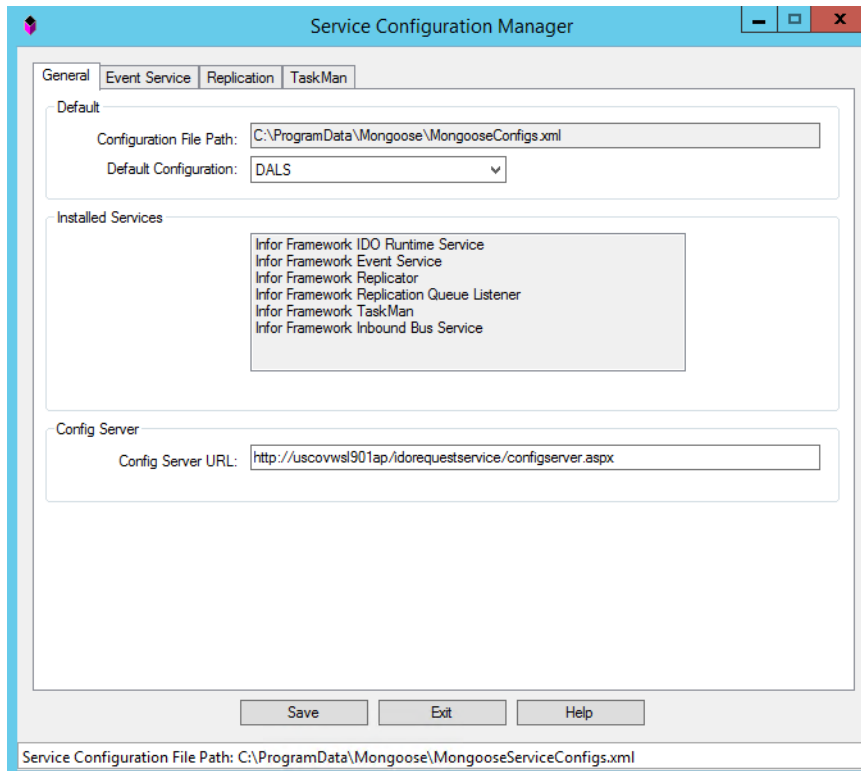
The Replication Tool can be used to view errors in any XML being sent from Infor Mongoose to a different Mongoose intranet or an external system. For most operating systems, navigate to **All Programs > Infor > Tools > Replication Tool**. For Windows Server 2012, find Replication Tool under Apps.



The Configuration and Listener queues are shown in the drop-down lists. Select them as the default to use in the other windows of this utility.

Service Configuration Manager (Setup)

Before you run the Replication Tool, use the Service Configuration Manager utility to configure replication options on the **Replication** tab. Then click **Save** to save these settings in the service configuration XML file specified on the **General** tab.



Instructions for using the Replication Tool and Service Configuration Manager are found in their online help.

Replication Errors Form (Inbound Flow)

During non-transactional replication, if an XML document makes it into an Mongoose intranet, is retrieved from the inbound MSMQ by the Replication Queue Listener, but fails when executed against the target site, its errors are displayed in Infor Mongoose's Replication Errors form on the target site. Generally these documents contain valid login and target site information but are failing for another reason.

If an inbound XML document fails before that point, you can view and correct it through the Replication Tool described above.

Use the form to fix and resubmit the data. If the update succeeds this time, the data is removed from the ShadowValuesErrors table; if it fails again, it will remain in the table.

The fields of interest are:

- **From Site:** Site where the record originated.
- **Object Name/Object Type:** Name and type of record to be replicated

- **Operation Number:** Distinguishes different elements in the errors.
- **Operation Type:** Type of operation: insert, update, delete, or method call.
- **Line:** Indicates which line of data for a particular record contained the error. The data is stored in rows of 55 name/value pairs, so for example, if there are 85 columns of data, there will be 2 lines.
- **Old/New:** Indicates whether a particular row of the collection represents the old or new values. For updates, the old values are also displayed.
- **Error:** Text of the error message received when this record was submitted.
- **Label:** Label seen on forms.
- **Value:** Data for a particular column. For example, the column "Description" might have data "bike seat" as a value.

Find Replication Setup Issue

If you know that a specific object, for example, a table, is not replicating changes from the current site to another site, use this form to help troubleshoot why replication is not occurring.

You enter an object name and type (a table, stored procedure, or XML document) and select the remote site name. When you click **List Issues**, any setup errors for that site/object combination are listed.

	Message
1	Replication Category that has [Replication Category Object: no_table] does not exist.
2	Replication Rule that has [Replication Category Object: no_table] and [From Site: Demo] and [To Site: MI] does not exist.
3	Intranet that has [From Site: Demo] does not exist.
4	Intranet that has [To Site: MI] does not exist.
5	User Name that has [From Site: Demo] and [To Site: MI] does not exist.

Some of the setup problems that may be displayed include:

- The From Site or To Site does not exist in the site table.
- No replication category contains the specified object (table, SP, etc.).
- No replication rule exists for the specified From Site and To Site, and for any category that contains the specified object.
- A rule exists for a category using the specified object, but the rule is currently disabled on the Replication Rules form.
- Replication is currently disabled between the specified From Site and To Site on the Sites or Sites/Entities form.

- No replication trigger exists for the specified object (if it is a table), or the trigger is disabled.
- The To Site has not been set up with a linked server name on the Sites or Sites/Entities form or through the Infor Mongoose Configuration Wizard. (Transactional replication)

Note: Transactional replication checks are made only if at least one transactional replication rule is enabled at this site.

- The To and From sites are on different database servers, and the To Site's database server has no link to it from this site's database server. This is established through the Linked Servers node in SQL Server. (Transactional replication)
- The To Site's database name is not specified on the Sites or Sites/Entities form. (Transactional replication)
- The user name specified for this From Site and To Site does not exist. (Non-Transactional replication)
- The replication user name is not a super user or has not been given access to the proper license module and group. (Non-Transactional replication)
- The To Site and/or From Site does not have an Intranet defined on the Sites or Sites/Entities form. (Non-Transactional replication)
- The To Site's intranet has no URL defined on the Intranets form. This is a problem only if the To Site is on a different intranet than the From Site. (Non-Transactional replication)

If no problems are found, the replication triggers may just need to be regenerated.

Troubleshooting Other Setup Problems

You may encounter other error messages during implementation. One common message and its solution is described below.

Message: Maximum Number of Databases Used for Each Query has been Exceeded

Close and restart the Infor Mongoose session where the error occurred. Retry the procedure you were running at the time of the error.

If you still get the error, close all open Infor Mongoose sessions and restart only the one for the configuration you were in when the error occurred. Retry the procedure that caused the error.

Framework Logging

All framework logging, from sources such as WinStudio, TaskMan, Replicator, and the IDO Runtime Service, is sent to a common logging routine. This log information can be viewed on the utility server through the Log Monitor utility, found in the program/app area under **Infor>Tools>Log Monitor**.

Note that information sent to the logs while the Log Monitor program is not running is not stored anywhere. If a developer is using the IDO Runtime Development Server instead of the Log Monitor, that program displays logging information.

Appendix B: Setting Up a Test Environment

B

Note: Refer to the Support Web site for the latest information on requesting license keys for demo or pilot databases.

Before you "go live" with a multi-site environment, you probably want to set up a test environment, possibly with your company's data. Follow the steps below to set up a multi-site test environment with transactional replication:

1. Create one test site. You have 2 choices:
 - Create an initialized application database that contains a "demo" site with data provided by Infor. This is described in the installation guide.
 - If you have already set up a site using this version of Infor Mongoose with your data in a "test" single-site environment, you can use that site as your initial site.
2. Repeat step 1 to create more application databases with different site names.

Note: The demo site has an invoice length to 10, so all of your test sites must have an invoice length of 10 if you are using any demo sites in your multi-site test setup.
3. Create at least one forms database. If you use the same forms database for both your live and test environments, be aware that any form customizations you make are then applied to both environments.
4. Create at least one object database.
5. On the database server, run the Mongoose Configuration Wizard to link all the sites that you want in your multi-site environment.
6. On the utility server, use the Configuration Manager to create client configurations that point to the new sites.
7. On the utility server, use the Service Configuration Manager to set up monitoring of your new configurations for TaskMan, non-transactional replication, and the application event system.
8. Log into one site configuration on each application database as "sa" .
 - a. In the License Management form, enter the license for this database. If you need to request special licenses, see the section in the *Licensing Guide* on licensing Demo and Pilot databases.

- b. In the Sites/Entities form, make sure the application and forms database names are set for each site in this database.
 - c. Log out.
9. Log into each site and set up replication rules and/or shared tables between all the sites, as described in Chapter 5, "Setting Up Replication at Sites."

Appendix C: Data Load To/From Spreadsheets

C

Requirements for Loading Data From Spreadsheets

If you have many sites with similar data to be entered at each site, it may make sense to load the data into the system from a spreadsheet. The spreadsheet method only works well if you have smaller volumes of data. It is much more difficult to deal with errors in spreadsheets when you are trying to load a large number of records at once.

Note: Consider using Infor Consulting Services if you have larger volumes of data to load. Contact your Infor representative for more information.

- Make sure the columns in the spreadsheet exactly match the order of the grid in the form, and that the data is in the format required by the form. The best way to do this is to create several sample records in the Mongoose form and then export them to a spreadsheet. Then use that data as a template for the other rows. Or you may want to create a "demo" database and export the sample data to use as a template.

In the template, you may want to hide columns for data that comes from calculated or read-only fields so that you do not enter data into them accidentally. Do not delete these columns; otherwise the columns will not match up when you paste them back into Mongoose.

- Make sure any supporting data has already been loaded.

Exporting Data From a Form to a Spreadsheet

To quickly create a spreadsheet that contains all the rows in the current form's grid, including the header rows, use this process:

1. In the form's grid, select the rows (shift-click in the row numbers on the left):

The screenshot shows a web application window titled "Replication Rules". It contains a table with the following data:

	Source Site	Target Site	Category
1	CAN	BC	Ledger Consolidation
2	CAN	BC	Shared Currency
3	CAN	BC	Site Admin

To the right of the table is a details panel with the following fields:

- Source Site: CAN (dropdown menu)
- Target Site: BC (dropdown menu)
- Category: Site Admin (text field)
- Description: (empty text field)

Note: You can use Ctrl-2 to toggle between the grid view and the form view.

2. Type Ctrl-C to copy the rows to the clipboard.
3. Open a spreadsheet and paste the rows into it. Notice that some columns are empty and some are set to a different format than appears in the form (for example, 0's and 1's instead of checkboxes).

	A	B	C	D	E	F	G	H	I
1	SourceSite	TargetSite	ReplicationCategory	Description	IntervalType	Interval	StartIn	Disable	UpdateAll
2	CRP	USA	Site Admin		Transactional	0		0	0
3	CRP	USA	Ledger Consolidation		Transactional	0		0	0
4	CAN	BC	Ledger Consolidation		Transactional	0		0	0
5	CAN	BC	Shared Currency		Transactional	0		0	0
6	CAN	BC	Site Admin		Transactional	0		0	0

Caution: The **Actions > To Excel** menu option is also available, but it is intended for export of data to other applications, not for updating data that will be imported back into Mongoose. By default, this option saves the columns in the order they appear in the IDO collection, which may not be the same order that they appear in the grid. (However, you can choose to match the sequence of columns in the grid.) Also, be aware that cells with data that contains hard and soft returns might not work as expected with the **To Excel** option. In addition, the **To Excel** output might include cells that are hidden on the grid. Be careful to ensure that the order of the cells are the same as the grid if you plan to copy the cells back into the application.

Importing Data From a Spreadsheet into a Form

1. Make sure the columns in the spreadsheet are in the same order as the columns in the form's grid.
2. In the spreadsheet, select the rows (shift-click in the row numbers on the left).

	A	B	C	D	E	F	G	H	I
1	SourceSite	TargetSite	ReplicationCategory	Description	IntervalType	Interval	StartIn	Disable	UpdateAll
25	CAN	CRP	Site Admin		Immediate	0		0	0
26	CAN	CRP	Ledger Consolidation		Immediate	0		0	0
27	CAN	USA	Site Admin		Immediate	0		0	0
34	CAN	ILL	Site Admin		Immediate	0		0	0
35	CAN	CAL	Site Admin		Immediate	0		0	0

3. Right-click and select **Copy**.
4. Make sure the form is in "Add" mode. that is, the **Create a New Object** toolbar button is enabled.

5. In the form's grid, click in the left column of the first empty row:

The screenshot shows the 'Replication Rules' form. On the left is a grid with columns for 'Source Site' and 'Target Site'. The first 8 rows are populated with data: (1, CAL, CAN), (2, CAL, CRP), (3, CAL, CRP), (4, CAL, ILL), (5, CAL, ILL), (6, CAL, ILL), (7, CAL, ILL), and (8, CAL, ILL). Row 9 is the first empty row, and its left column is selected. To the right of the grid are configuration fields: 'Source Site' (dropdown), 'Target Site' (dropdown), 'Category' (text field), 'Description' (text field), 'Interval Type' (dropdown set to 'Immediate'), 'Interval' (text field), 'Start Interval At' (text field), and two checkboxes: 'Disable Replication' and 'Update All Columns'.

6. Select **Edit>Paste Rows Append**. The rows are added in the grid:

The screenshot shows the 'Replication Rules' form after pasting rows. The grid now contains 11 rows. The first 8 rows are the same as in the previous screenshot. Rows 9, 10, and 11 are new: (9, CAN, CRP), (10, CAN, USA), and (11, CAN, ILL). The left column of row 11 is selected. The configuration fields on the right are updated: 'Source Site' is 'CAN', 'Target Site' is 'ILL', 'Category' is 'Invoice Builder', and 'Interval Type' is 'Immediate'. The 'Interval' and 'Start Interval At' fields are empty, and the checkboxes remain unchecked.

7. Click the Save button to save the rows.

The copy works only if all the information in the rows is valid and in the proper format. Otherwise, you get data validation errors and the import stops. See “Requirements for Loading Data From Spreadsheets” on page 73.

When the system encounters an error, it asks if you want to correct the field. If you answer **Yes**, the import stops. If you answer **No**, the system continues pasting the rest of the rows. When it completes, you can correct the rows marked with "ne" in the Mongoose form.

Appendix D: Removing a Site from an Existing System

D

Note: Be aware that, if you remove a master site, intranet licensing will no longer be available, since these features depend on having a master site.

The steps below describe removal of a site that exists by itself in an application database.

For more information on any step, refer to the online help.

1. Make sure all other users are logged out of the site. If you are removing a master site, all users must be logged out of all sites on the master sites' intranet. If the remaining sites are all in one application database, and the removed site is the *only site* not in that database, all users must be logged out of that application database.
2. If the site belongs to an intranet that is sharing tables:
 - a. Go to the *master site's* Intranet Shared Tables form and unshare all the shared _all tables. This adds the shared tables back to all of the sites in the intranet.
 - b. Go to the *master site's* Intranet Shared User Tables form and unshare the shared user tables. This adds the shared user tables back to all of the sites in the intranet.
 - c. At the *site being removed*, go to the Sites or Sites/Entities form and change the Intranet value to some other intranet (or make it blank).
 - d. If the site being removed **is** the *master site for the shared intranet*, log into the *master site*. On the Intranets form, blank out the Master Site field. This should replicate to all the other sites in the shared intranet.
3. If the site being removed uses intranet licensing, deselect the **Intranet Licensing** check box on the Sites or Sites/Entities form and save the record.

If the site being removed is a *master site* using intranet licensing, you must also deselect the **Intranet Licensing** field at all slave sites and then apply a license at each site.

Caution: Turning Intranet Licensing off at a site requires that a license document be applied to that site. It may also result in incomplete licensing on that site for some user-created objects, because the ModuleMembers records created for new and copied forms will have been created in the master site application database and in the originator's site application database. If Intranet Licensing at a site is disabled, and if it is not the master site or the site from which the new or copy action was originated, then the appropriate ModuleMember licensing records for those user-created forms and IDOs are not present in the application database of that site

Users cannot access sites where you have turned off intranet licensing until you either set up specific licensing for that site, or set up a new master site with intranet licensing and then reselect the **Intranet Licensing** check box at the master and slave sites.

4. Remove replication rules:
 - from other sites to the site being removed
 - from the site being removed to other sites.
5. If the removed site uses a linked server definition to connect to other sites (see step 9 on page 13), change the linked server setup within SQL Server or with the Configuration Wizard on the database server.
6. If the remaining sites are all in one application database, and the removed site is the *only site* not in that database, use the **Update _All Tables** form to truncate the _all tables in each site.
7. Delete the removed site's record in the Sites or Sites/Entities form at the other sites.

Since the Site Admin category is replicated between all sites, removing the site record from one site should remove it from all of the others. Verify this at each site.

You may need to delete data relating to the removed site, so that Infor Mongoose will allow the deletion of the removed site's records on the Sites or Sites/Entities form. If so, an error message displays information about the problem. You can also use SQL queries to review any data that references the site, to make sure you are not leaving databases in a state where orphaned data causes issues with functionality.

Some records for the site are automatically removed by the system. For example, the site's record is removed from the Site Groups form.

Site records remaining in _all tables should not cause problems with the application. However, they do take up space in the database. See step 13.

8. If the remaining sites are all in one application database, and the removed site is the *only site* not in that database, use the **View Management** and **Trigger Management** forms to regenerate the views and table triggers in that application database.
9. If the site being removed was the master site, specify a new master site (optional). Log into the site that will be the new master site. On the Intranets form, select the shared intranet. In the **Master Site** field, select this site to specify it as the master site for the intranet.
10. At the *master site*, delete any references to the removed site in the Sites or Sites/Entities form's **Link Info** tab.

11. In the *removed site*, delete records for the other sites in the Sites or Sites/Entities form. See the Note above.
12. Regenerate the replication triggers at *all sites*.
13. You may want to use the Update _All Tables utility to truncate and repopulate the _all tables, deleting references to the removed site. Or you may want to temporarily preserve the site records in the _all tables for later auditing. If you choose to truncate/repopulate, perform the truncation at all sites first; then perform the repopulation at all sites. (If you truncated the tables in step 9, you can skip this step.)
14. If the site was removed from an intranet with shared _all tables, go to the *master site*. Open the Intranet Shared Tables form and re-share the shared _all tables that you unshared in step 2.
15. If the site was removed from an intranet with shared user tables, go to the *master site*. Open the Intranet Shared User Tables form and re-share the shared user tables that you unshared in step 5.
16. In the Service Configuration Manager on the utility server, if the removed site's configurations were being monitored by the Replication/Inbound Bus services, or by TaskMan or the Event Service, remove the configurations from the list of monitored configurations.
17. In the Configuration Manager on the utility server, remove the site's configurations from any configuration groups, and remove or edit the configuration definitions.
18. Restart the Infor Framework services on the utility server to pick up the configuration changes.

After removing this site from its current system, you might want to add it to another Mongoose multi-site system.

Appendix E: Upgrading a Multi-Site System

A red square containing a white capital letter 'E'.

To upgrade from Mongoose version 7 or 8, see the *Infor Mongoose Installation Guide* chapter on upgrading an Mongoose database. Follow the steps in that chapter to upgrade each of your application databases.

Note: If you plan to keep all of your sites in separate databases, then stop here and do not continue with the steps below.

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